

Basics of vision II: The brain

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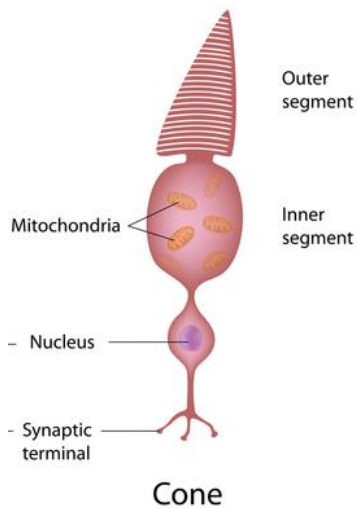
Revision and catch-up from Tuesday

Rods vs cones: 1) number

Differences between photoreceptors

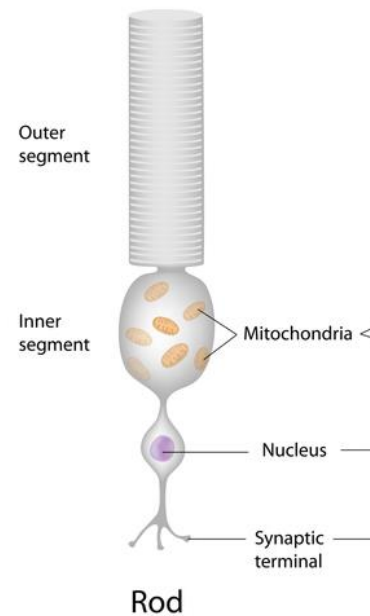
Cones

6m/eye



Rods

120m/eye



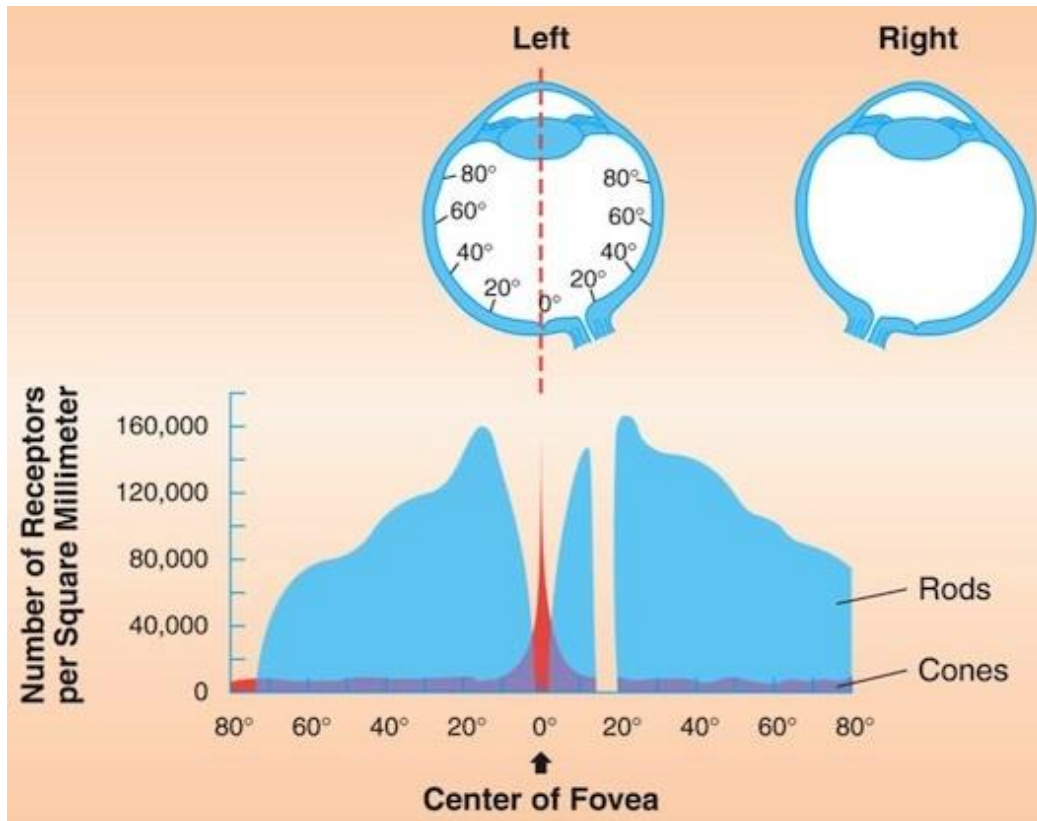
Rods vs cones: 2) distribution

Cones

Fovea & periphery

Rods

Retinal periphery



Fovea:

Cones only

Retinal periphery:

More rods than cones, density decreases with increasing distance from fovea.

Rods vs cones: 3) dark-adaptation

Cones

Fovea & periphery

Light-adapted, operates at high luminance

Rods

Retinal periphery

Dark-adapted, operates at low luminance

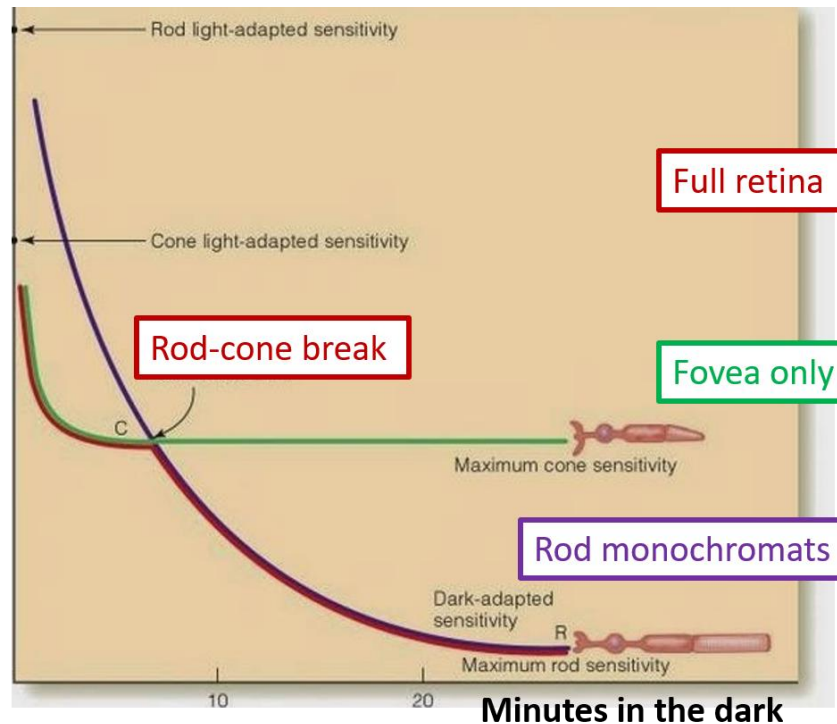
Sensitivity to light

LOW

(**Day**: a lot of light needed for sensation)

(**Night**: little light needed for sensation)

HIGH



Complete **dark adaptation** (i.e., ability to detect a light in the dark) is achieved by the rods, because they are more sensitive to light.

Rods vs cones: 5) acuity

Cones

Fovea & periphery

Light-adapted, operates at high luminance

Low light sensitivity

High visual acuity

Rods

Retinal periphery

Dark-adapted, operates at low luminance

High light sensitivity

Low visual acuity

→ Both have to do with **neural convergence**.

Many neurons synapse onto **fewer** neurons.



120 mil rods
6 mil cones



1 mil ganglion cells

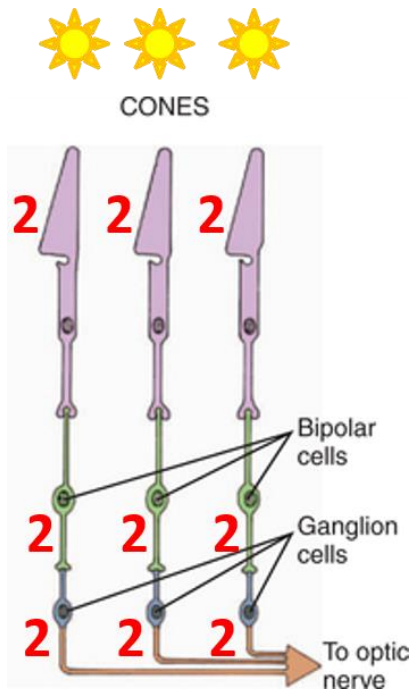
Rods vs cones: 4) light sensitivity

Cones

Fovea & periphery

Light-adapted, operates at high luminance

Low light sensitivity

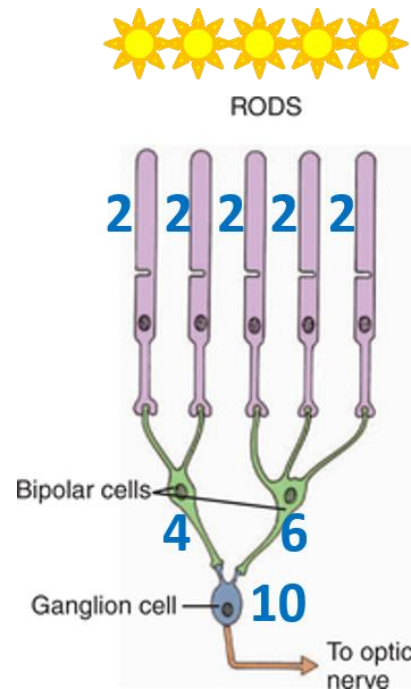


Rods

Retinal periphery

Dark-adapted, operates at low luminance

High light sensitivity



Neural convergence

(i.e., summed firing rates) increases the likelihood to detect a light stimulus (i.e., light sensitivity).

Rods vs cones: 5) acuity

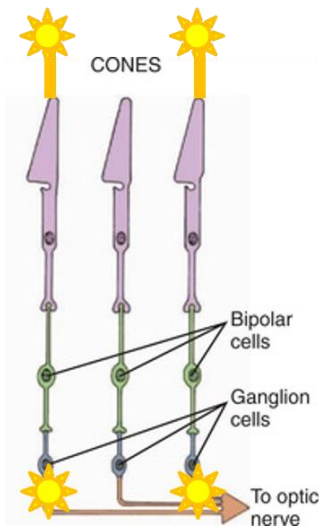
Cones

Fovea & periphery

Light-adapted, operates at high luminance

Low light sensitivity

High visual acuity



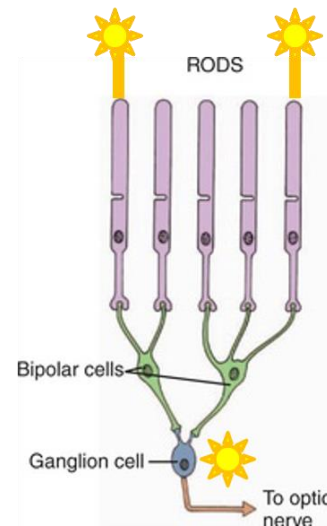
Rods

Retinal periphery

Dark-adapted, operates at low luminance

High light sensitivity

Low visual acuity



Neural

convergence

(i.e., summed firing rates) decreases the ability to locate a light stimulus (i.e., vision is less detailed).

Rods vs cones: 6) spectral sensitivity

Cones

Fovea & periphery

Light-adapted, operates at high luminance

Low light sensitivity

High visual acuity

Sensitive to long light waves

Rods

Retinal periphery

Dark-adapted, operates at low luminance

High light sensitivity

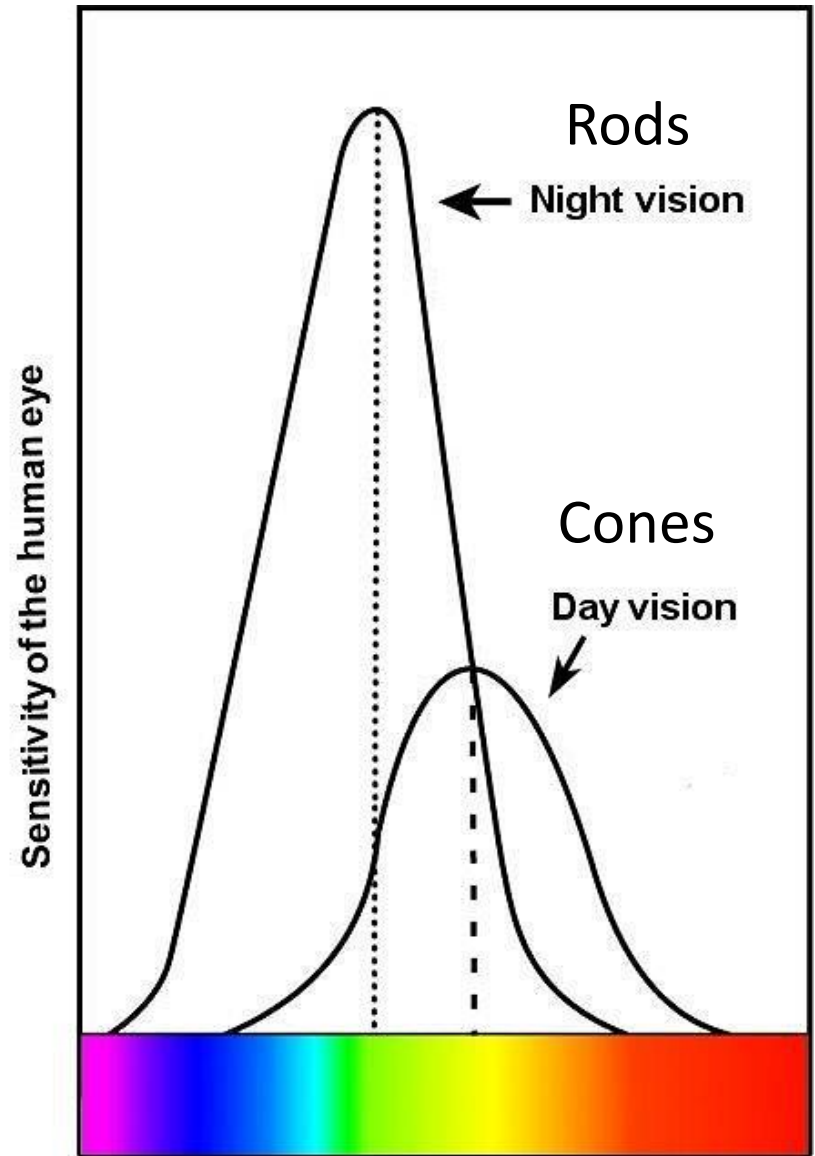
Low visual acuity

Sensitive to short light waves

Rods vs cones: 6) spectral sensitivity

6. Spectral sensitivity

- **Rods** are more sensitive to **shorter wavelengths**, maximum at 500 nm.
- **Cones** are more sensitive to **longer wavelengths**, maximum at 560 nm.

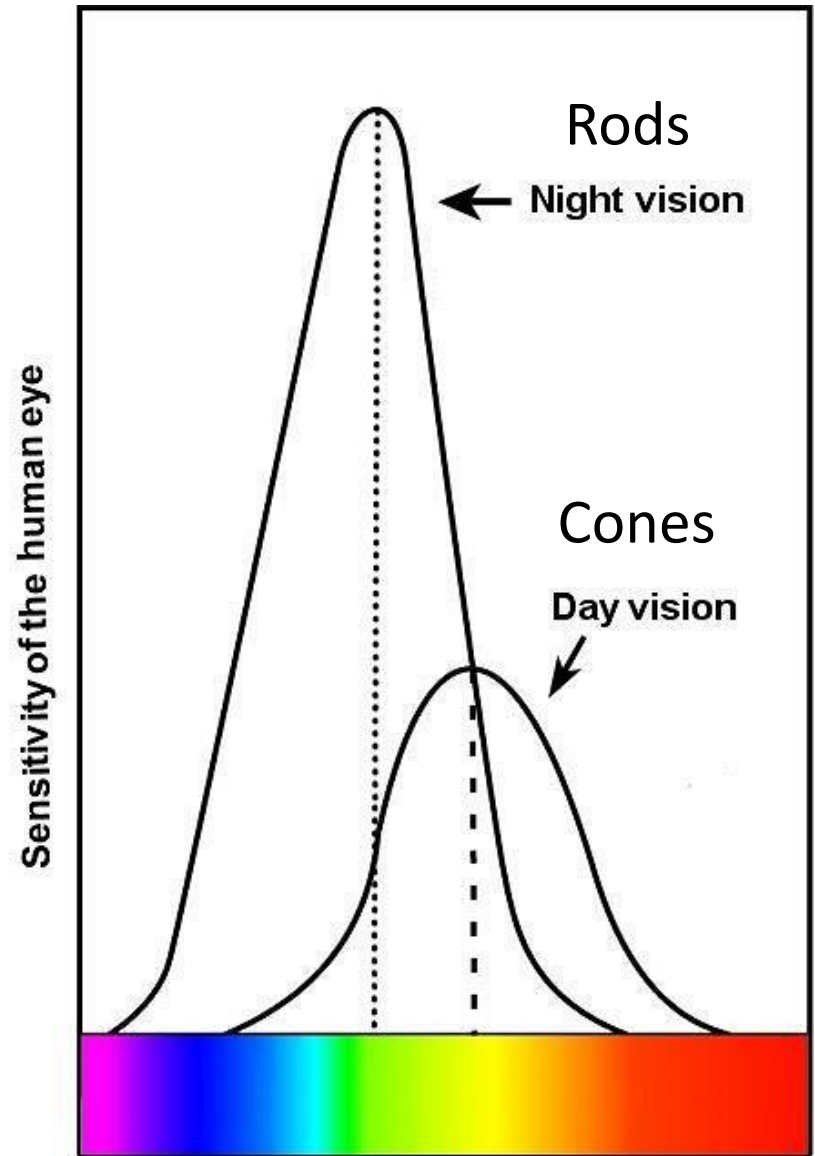


Rods vs cones: 6) spectral sensitivity

6. Spectral sensitivity

— Purkinje shift

Spectral sensitivity shifts from long to short waves in the light- versus dark-adapted eye.
(Johann Purkinje, 1825)

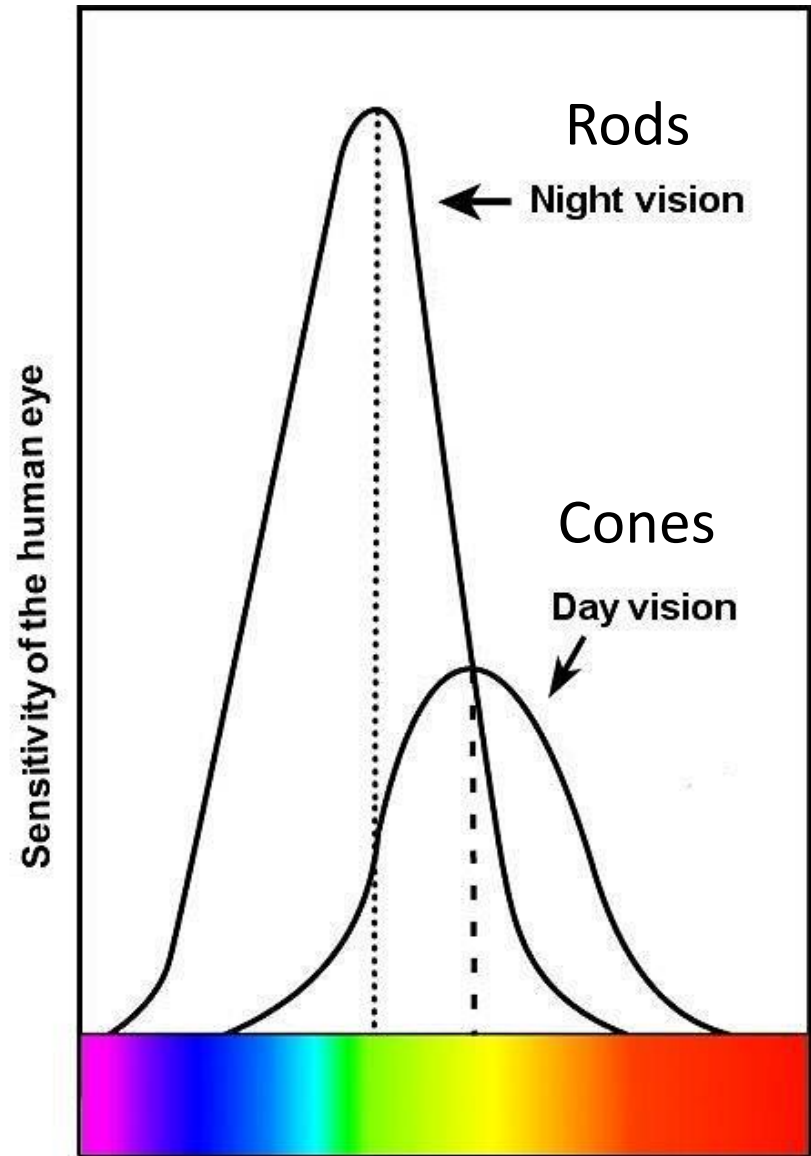
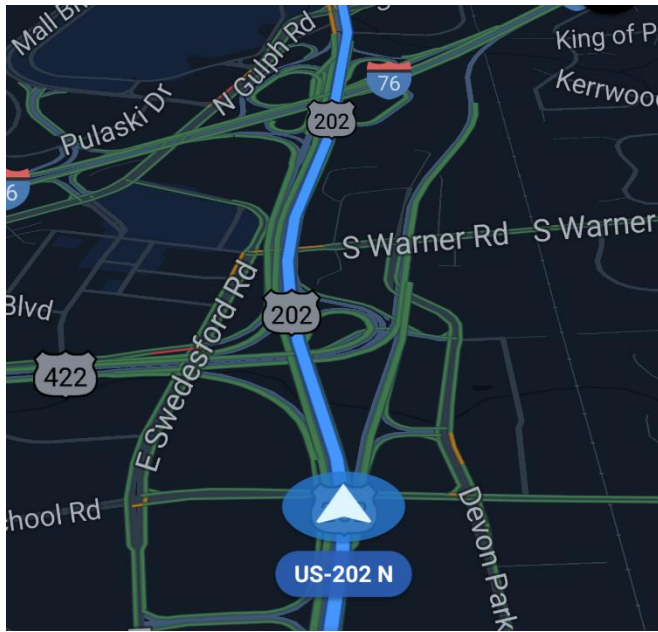


Rods vs cones: 6) spectral sensitivity

6. Spectral sensitivity

— Purkinje shift

Spectral sensitivity shifts from long to short waves in the light- versus dark-adapted eye.



Rods vs cones: 7) colour vision

Cones

Fovea & periphery

Light-adapted, operates at high luminance

Low light sensitivity

High visual acuity

Sensitive to long light waves

Basis of colour vision

Rods

Retinal periphery

Dark-adapted, operates at low luminance

High light sensitivity

Low visual acuity

Sensitive to short light waves

No colour sensation

Rods vs cones: 7) colour vision

7. Colour vision

- Rod vision = dark-adapted vision
operates at low luminance
No colour sensation
- Cone vision = light-adapted vision
operates at high luminance
S cones (respond to short wavelengths, blue)
M cones (respond to medium wavelengths, green)
L cones (respond to long wavelengths, red)



2 types of vision

Photopic vision

Cones

Fovea & periphery

Light-adapted, operates at high luminance

Low light sensitivity

High visual acuity

Sensitive to long light waves

Basis of colour vision

Scotopic vision

Rods

Retinal periphery

Dark-adapted, operates at low luminance

High light sensitivity

Low visual acuity

Sensitive to short light waves

No colour sensation

(**Mesopic vision** = rod and cone vision together)

The visual pathway: Last time we looked at...

2nd transformation: Transduction

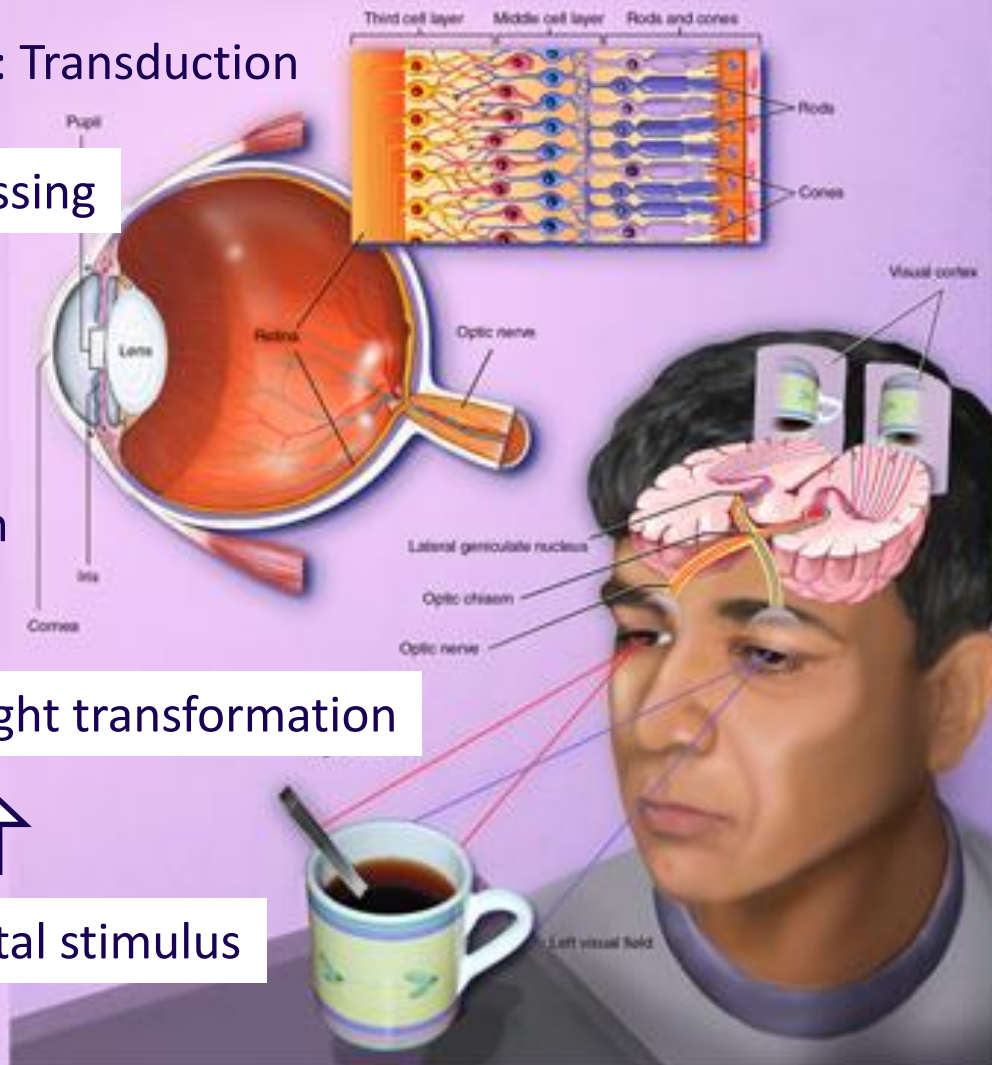
3. Receptor processing

1st transformation

Representation

2. Light transformation

1. Environmental stimulus



The visual pathway: Today we will look at...

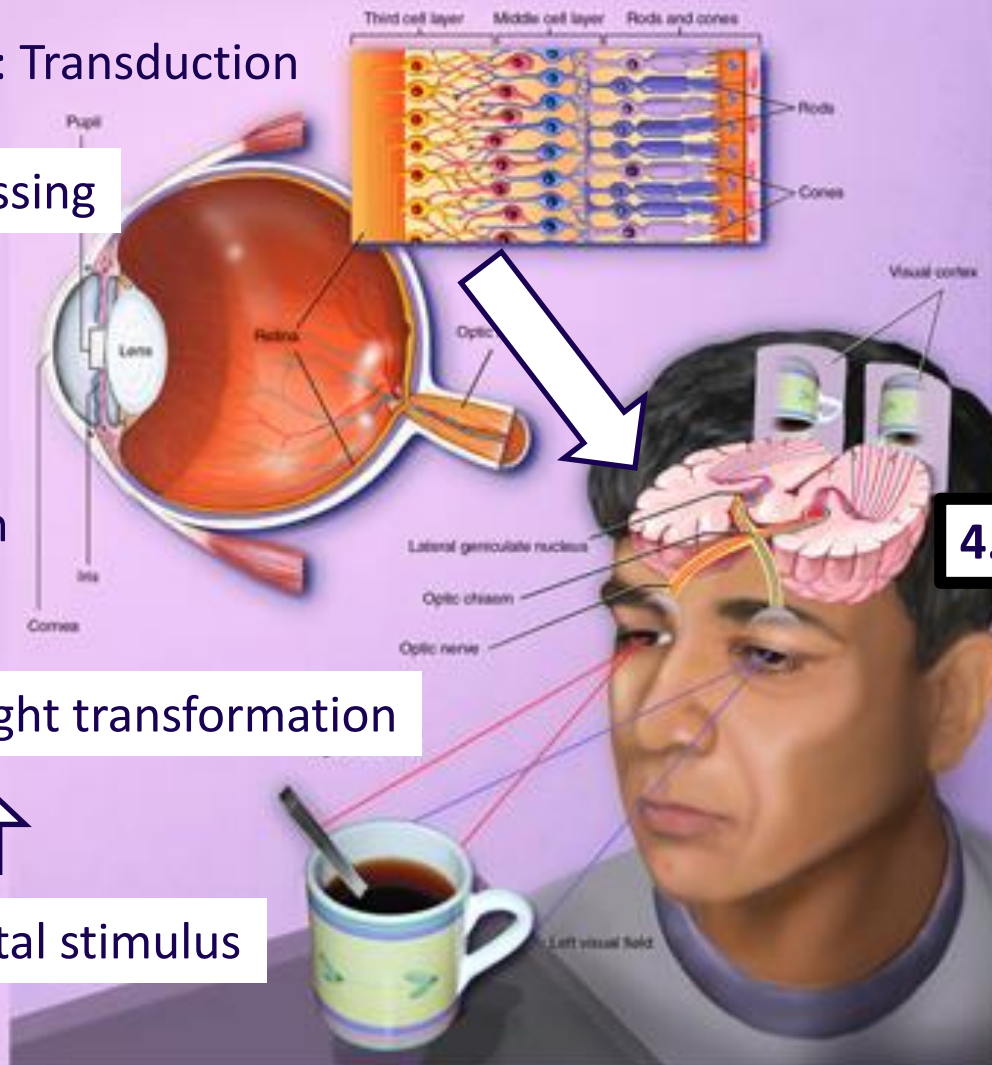
2nd transformation: Transduction

3. Receptor processing

1st transformation
Representation

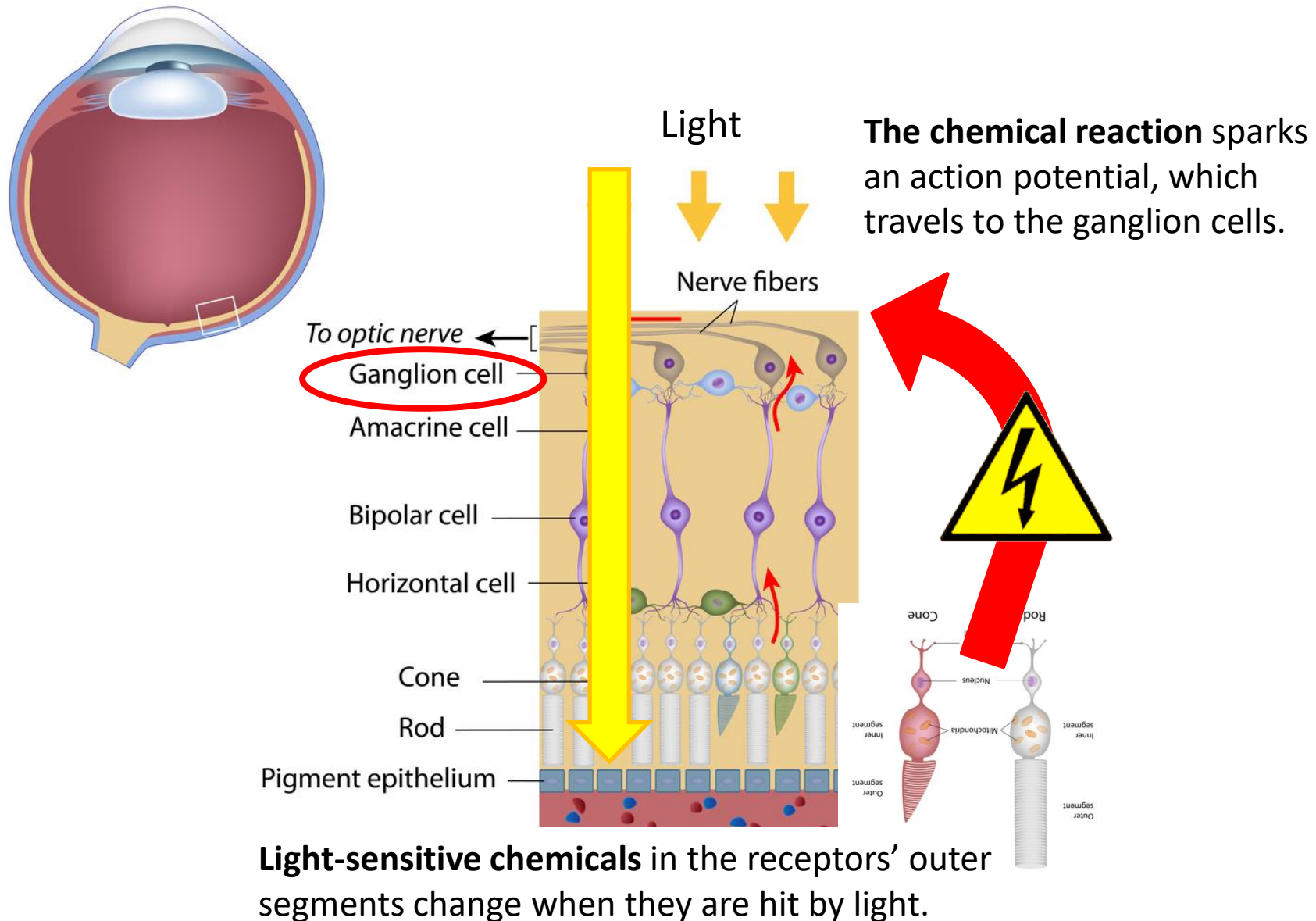
2. Light transformation

1. Environmental stimulus



4. Neural processing

The visual process: 2nd transformation



Ganglion cells

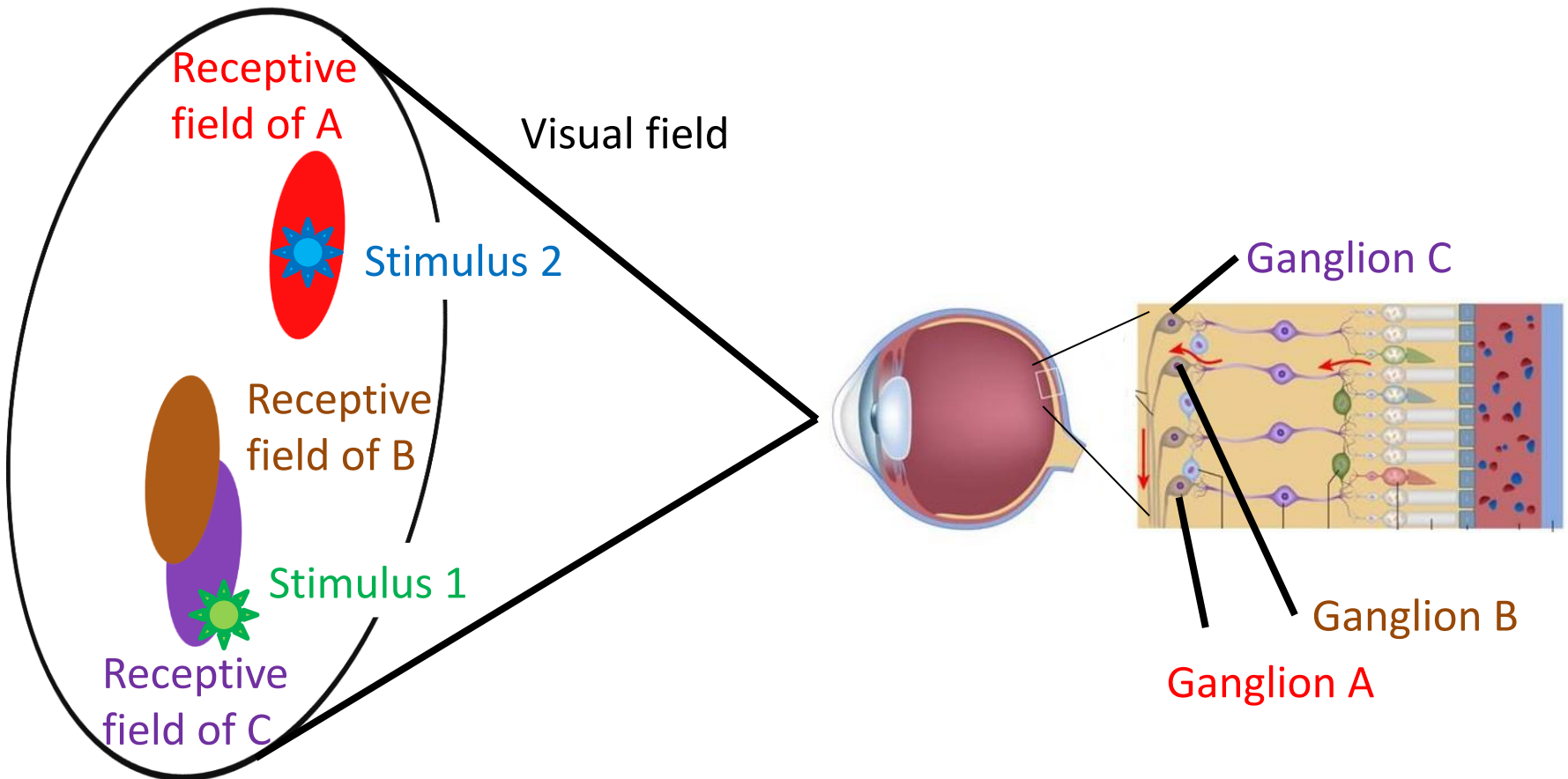
- Ganglion cell input
 - Neural Convergence

Each ganglion cell receives input from ~126 photo receptors.
Note variance: May be 1:1 in fovea but 1:100^x in periphery!
 - This input either **excites or inhibits** a ganglion cell...

...depending on where in a ganglion cell's **receptive field** the stimulus is located.

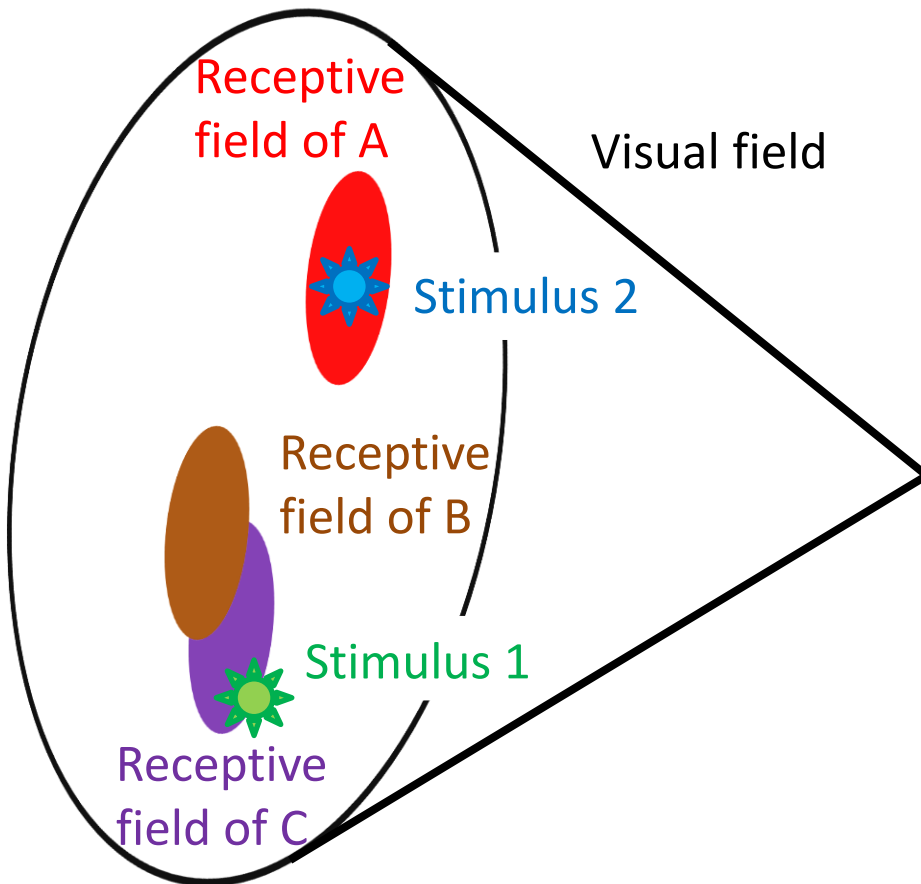
Ganglion cells

- **Receptive field?** The receptive field of a neuron is the **part of the visual field** in which a stimulus can modify the neuron's firing rate.



Ganglion cells

- Receptive field? The receptive field of a neuron is the **part of the visual field** in which a stimulus can modify the neuron's firing rate.

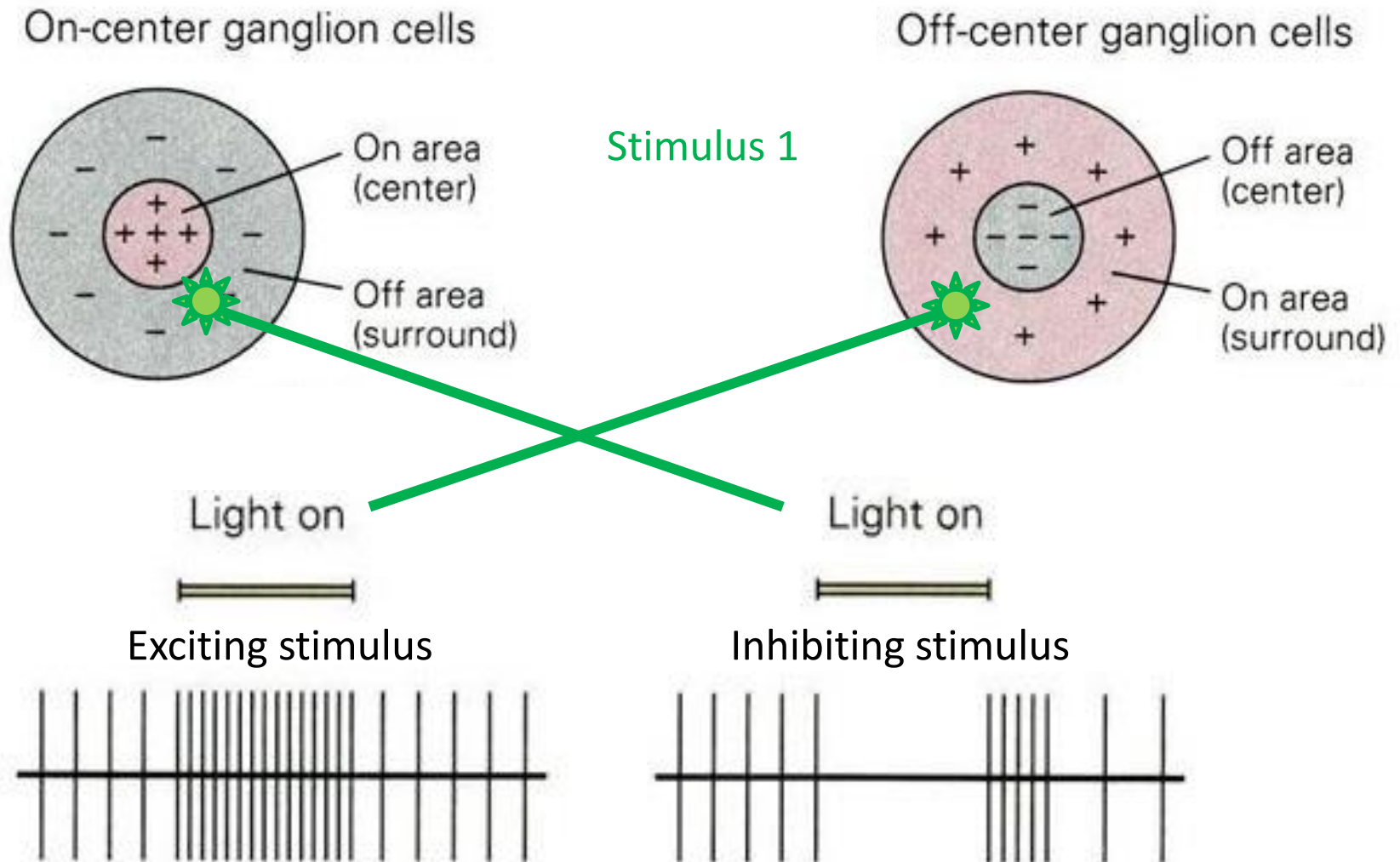


Does it matter whether a stimulus falls into the centre or periphery of a ganglion cell?

YES!

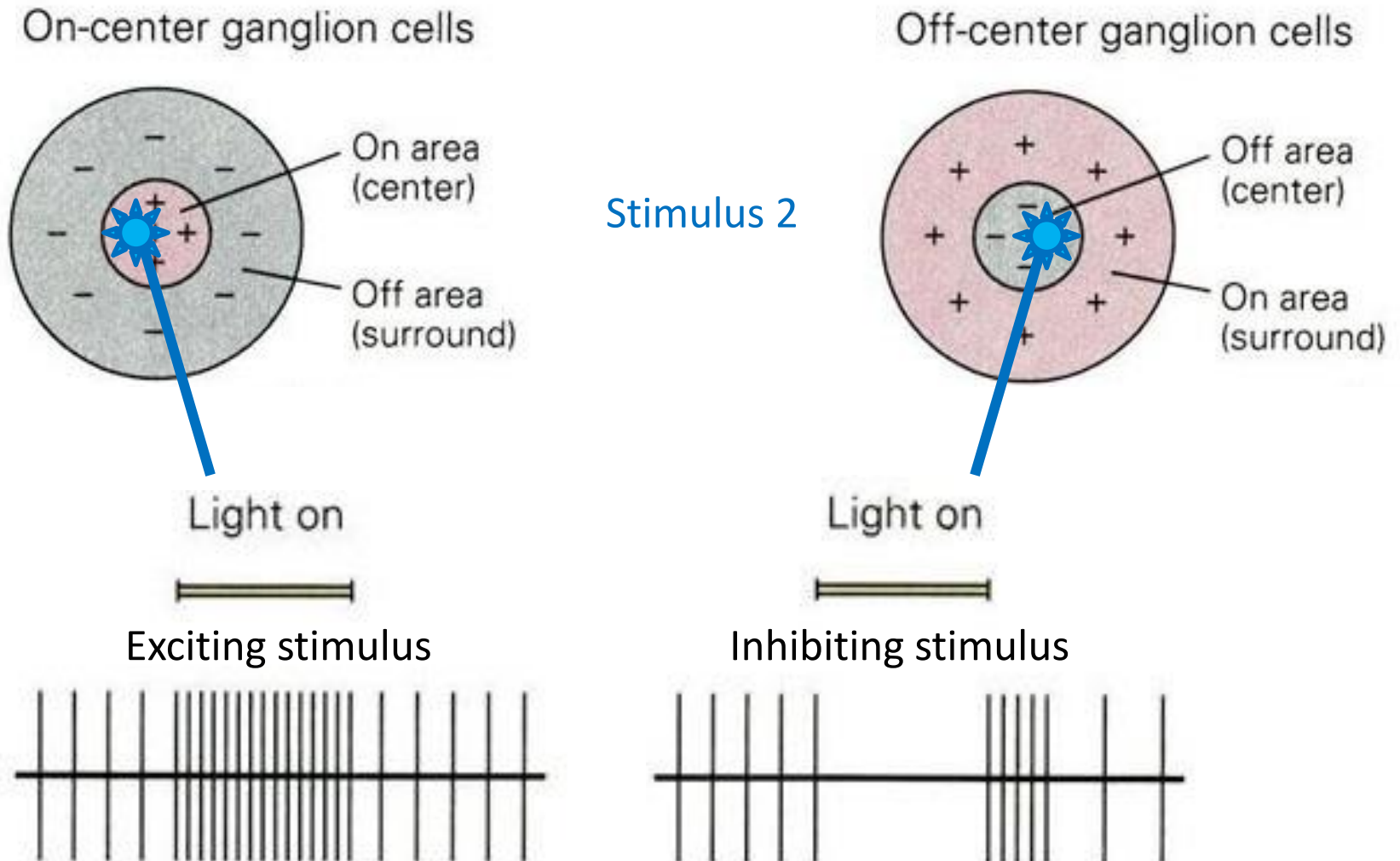
Ganglion cells

Ganglion cells have **on₍₊₎/off₍₋₎ centre/surround** receptive fields.



Ganglion cells

Ganglion cells have **on₍₊₎/off₍₋₎ centre/surround** receptive fields.



Ganglion cells

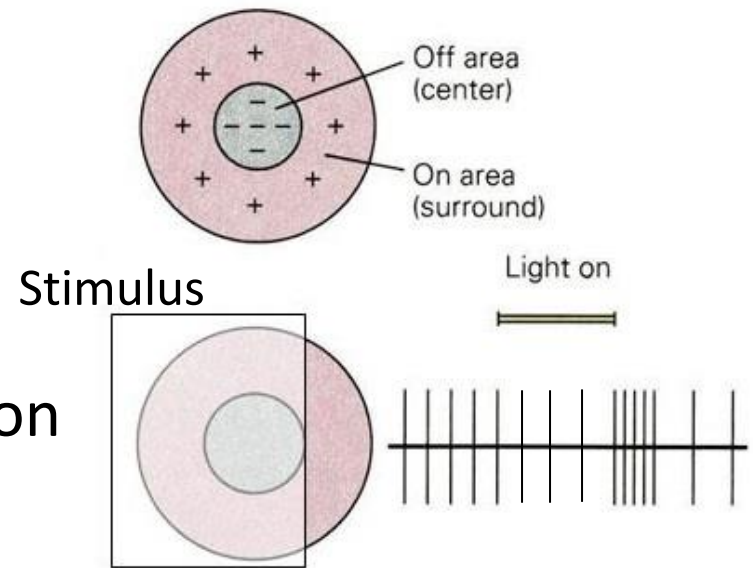
Ganglion cells have **on₍₊₎/off₍₋₎ centre/surround** receptive fields.

- Excitatory area (+)
 ‘on’ area
 increases the firing rate of a neuron when stimulated.
- Inhibitory area (-)
 ‘off’ area
 decreases the firing rate of a neuron when stimulated.

Ganglion cells

- Why is this important?

- Uniform illumination will not interest an on/off cell, activation cancels out.

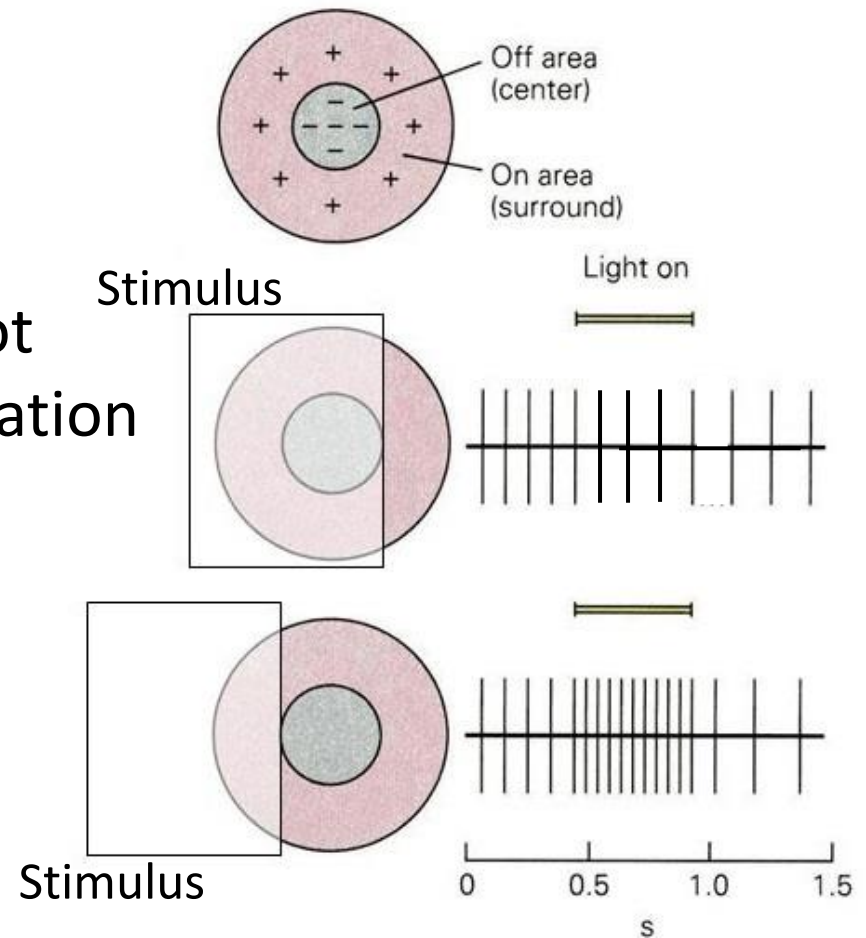


Ganglion cells

- Why is this important?

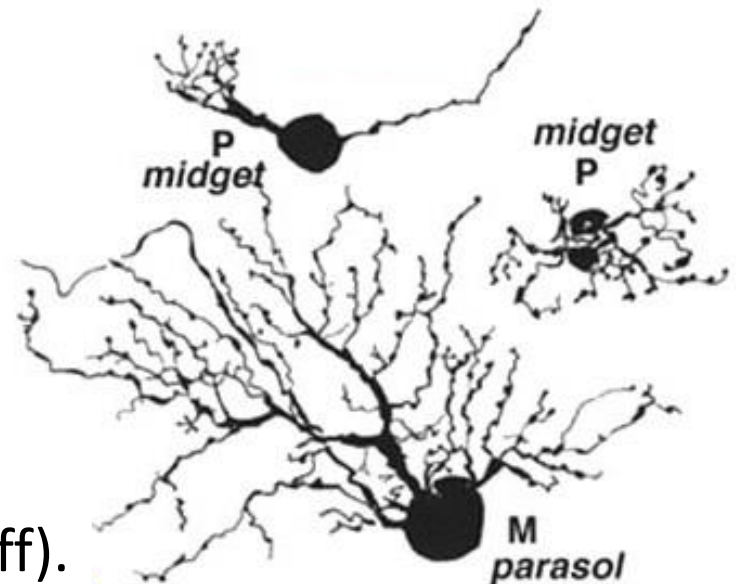
- Uniform illumination will not interest an on/off cell, activation cancels out.

- A **luminance discontinuity** (e.g., an edge of an object) causes activation.



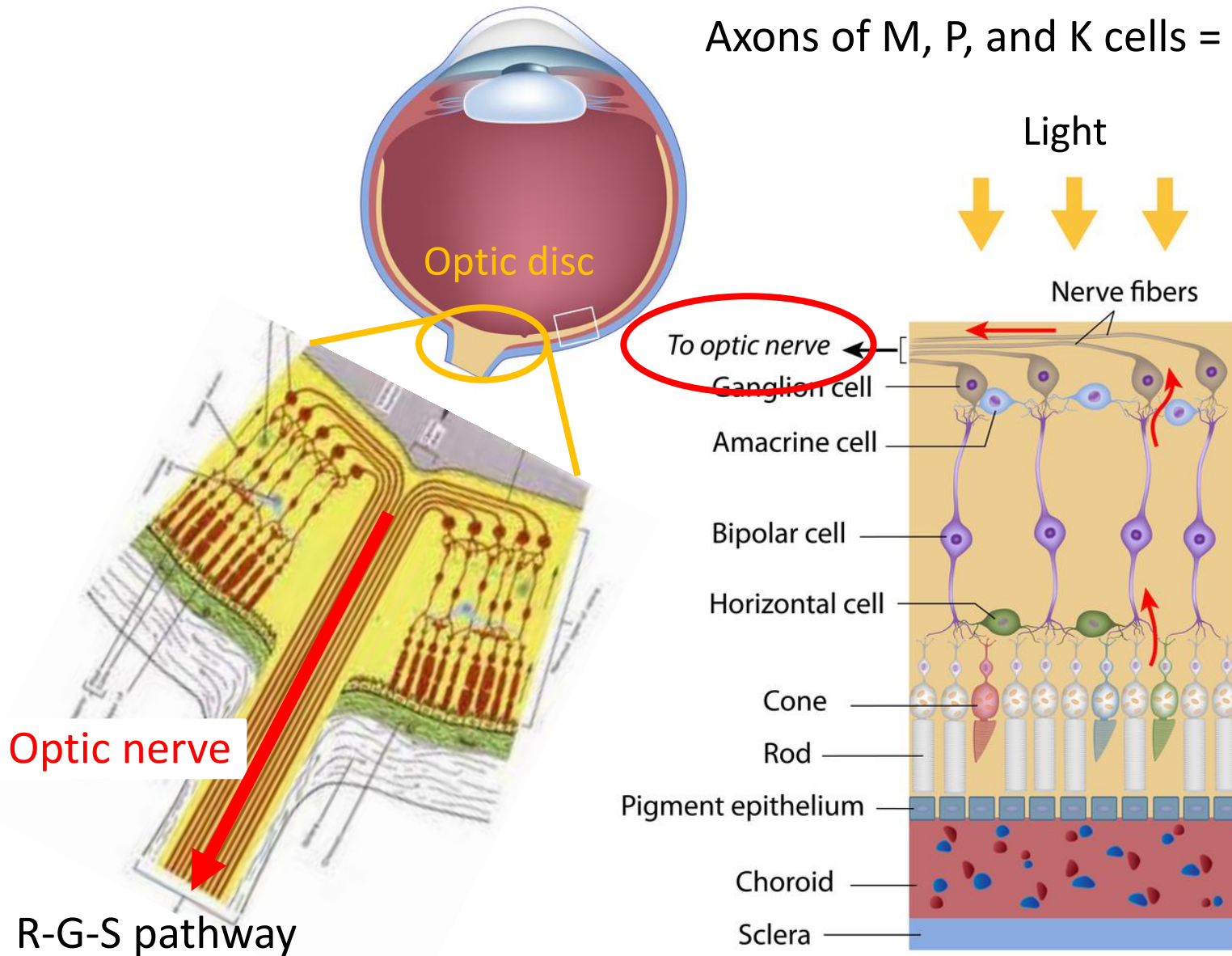
Ganglion cells

- 3 main types of ganglion cells
 - Magnocellular (M) cells.
Mostly input from **rods**.
Not colour specific.
 - Parvocellular (P) cells.
Input from single **M or L cones**.
Colour specific (green or red on/off).
 - Koniocellular (K) cells.
Excitatory input from **S cones**, inhibitory
input from M and L cones (blue on).

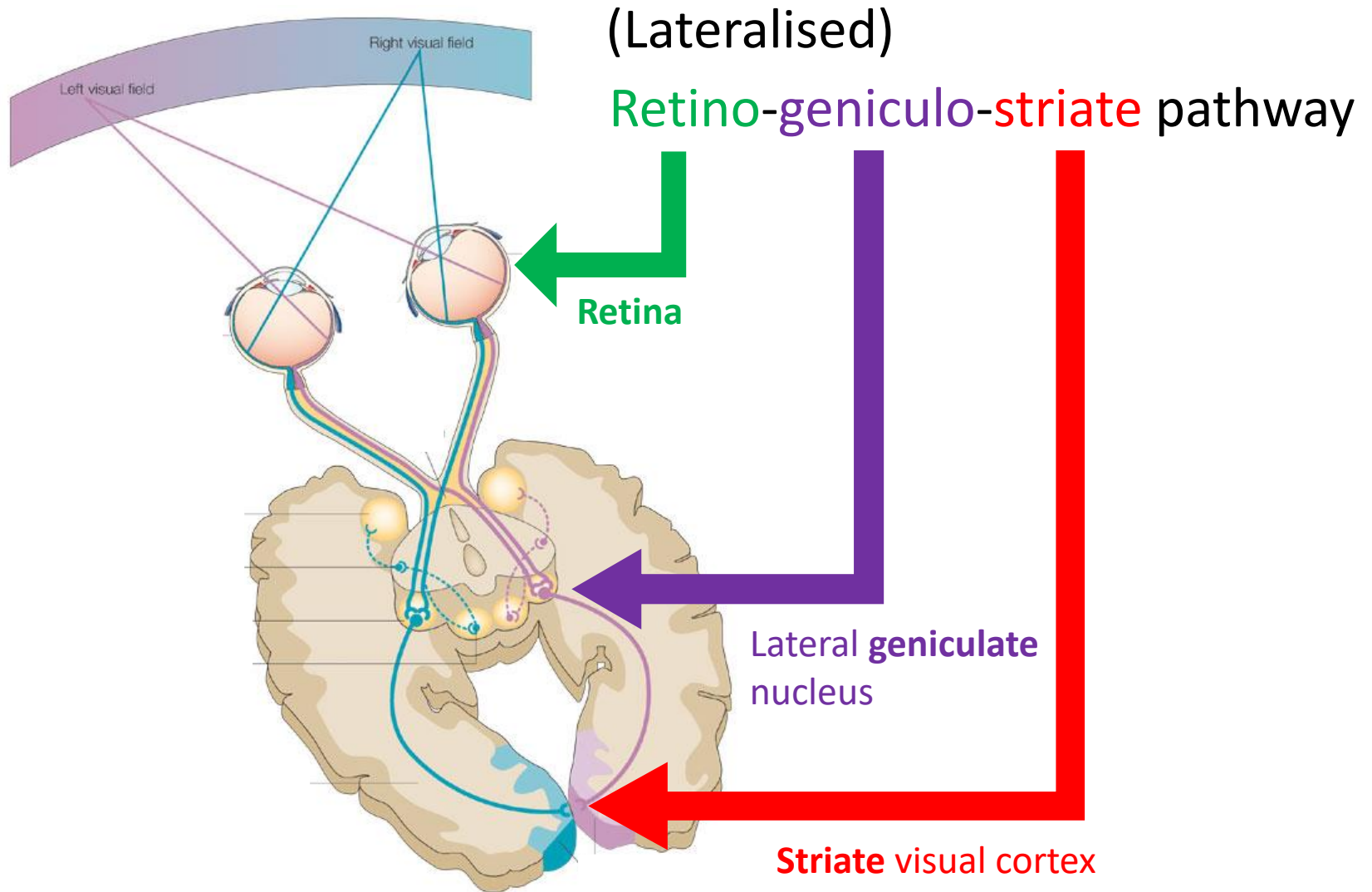


Ganglion cells

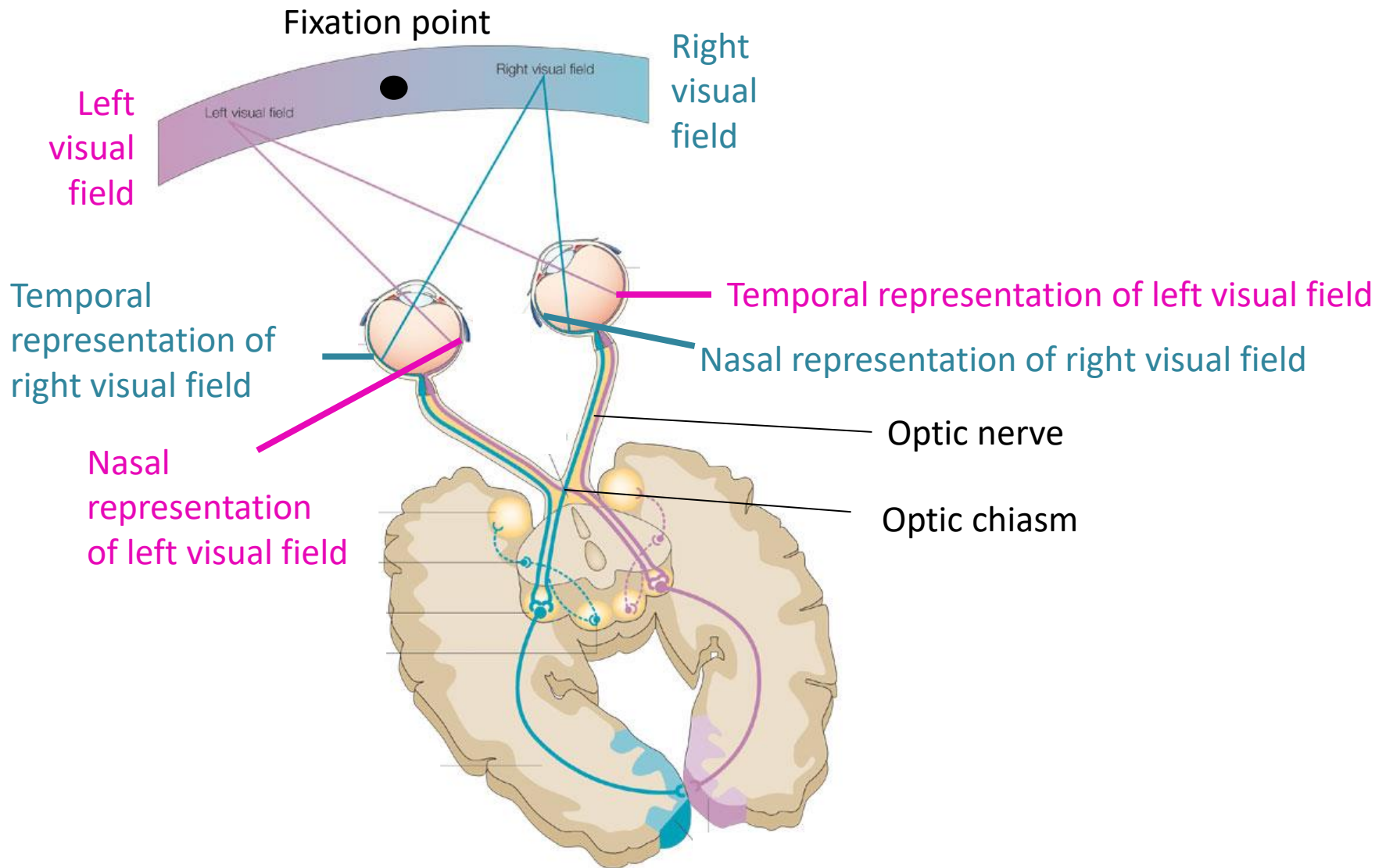
Axons of M, P, and K cells = optic nerve



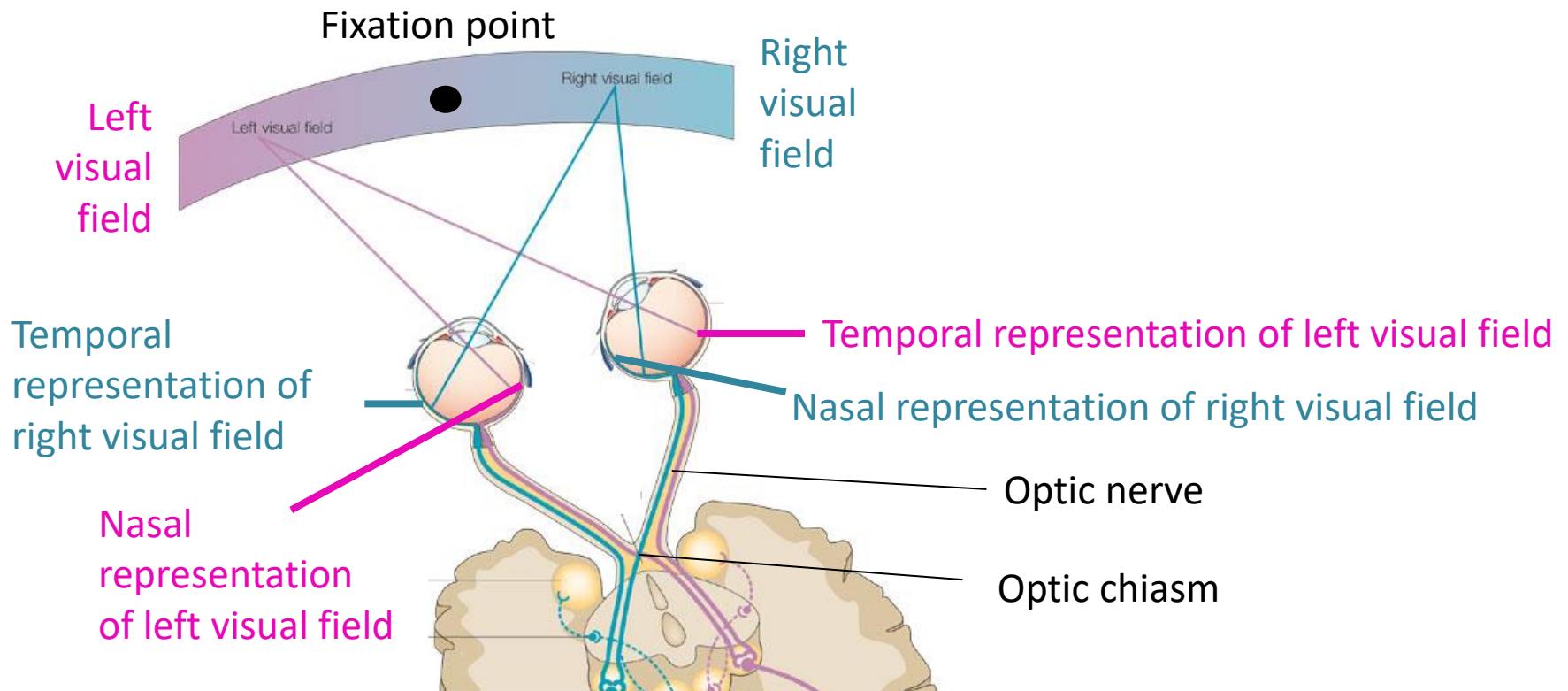
Retino-geniculo-striate pathway



Retino-geniculo-striate pathway



Retino-geniculo-striate pathway



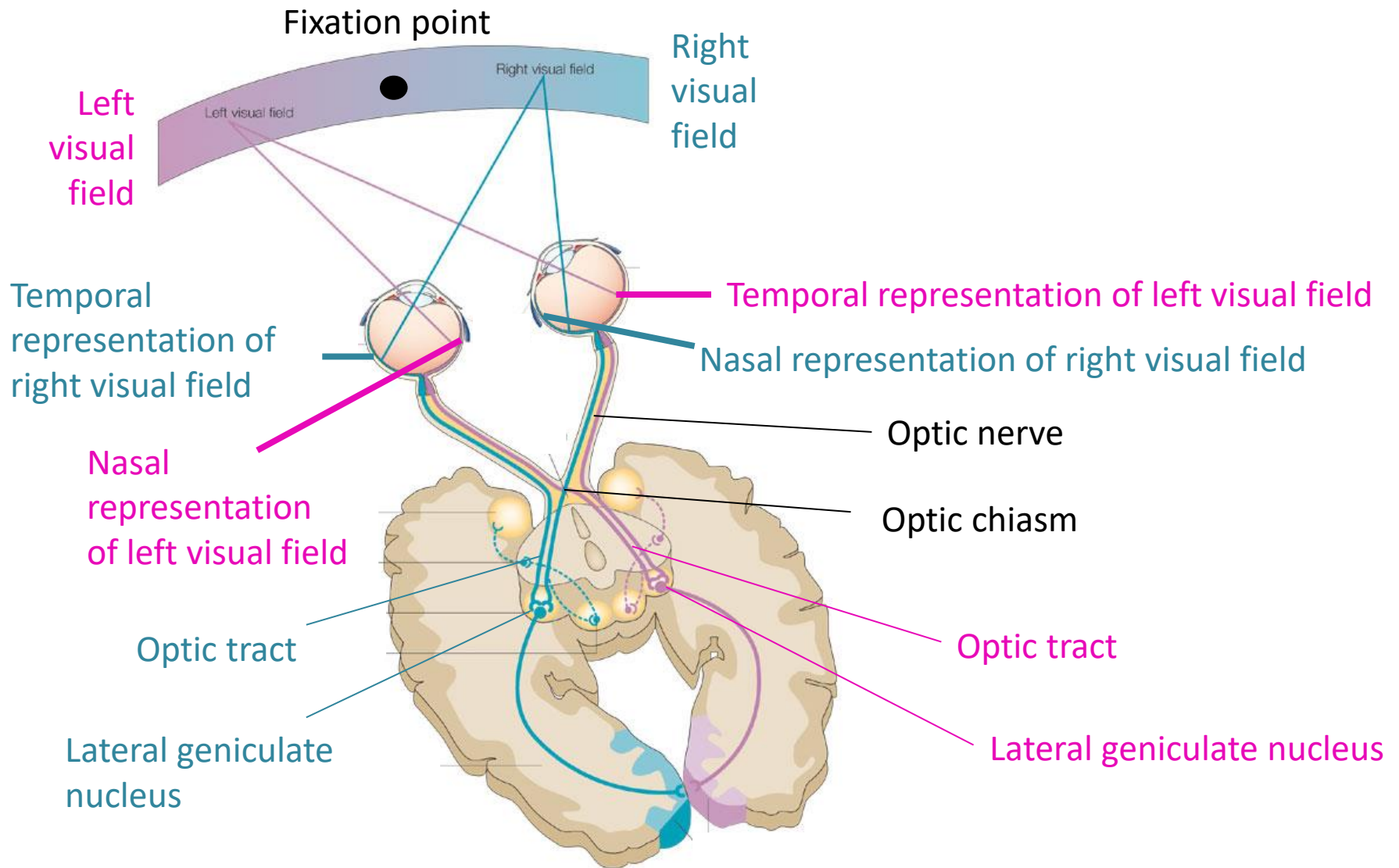
Lateralisation in optic chiasm

Nasal axons cross over to the other side of the brain

Temporal axons stay on the same side

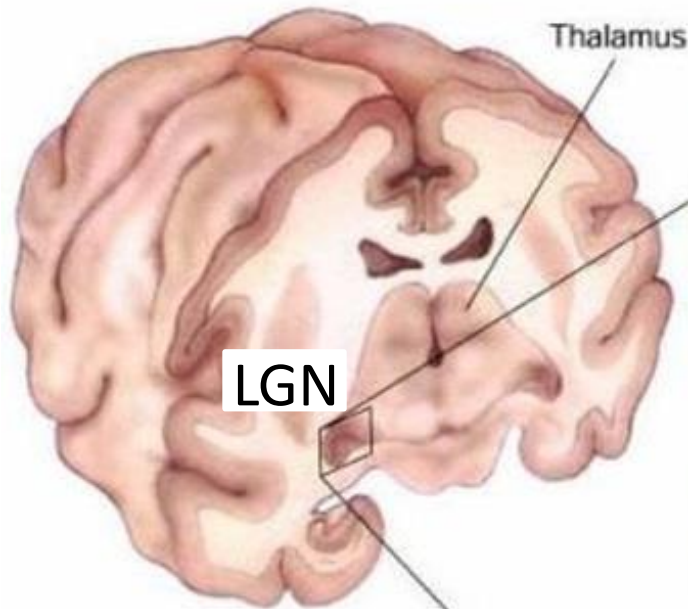
→ Visual fields are now represented in **contralateral hemispheres**
(left visual field in right hemisphere, right visual field in left hemisphere)

Retino-geniculo-striate pathway

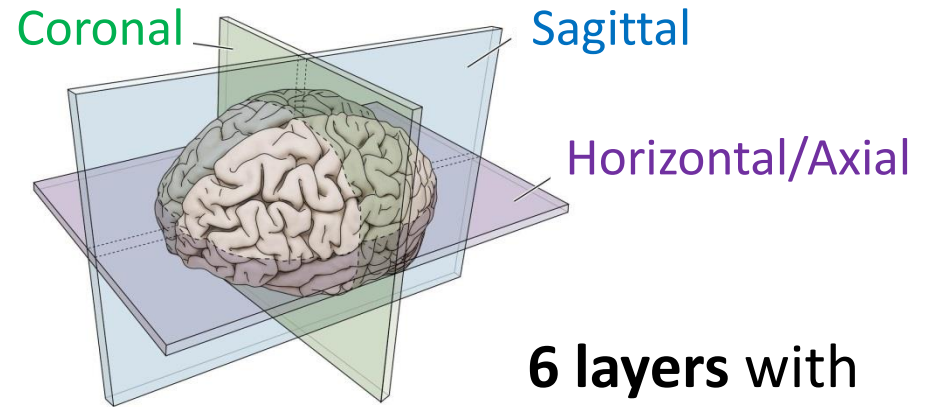


Lateral geniculate nucleus (LGN)

Coronal cut



LGN



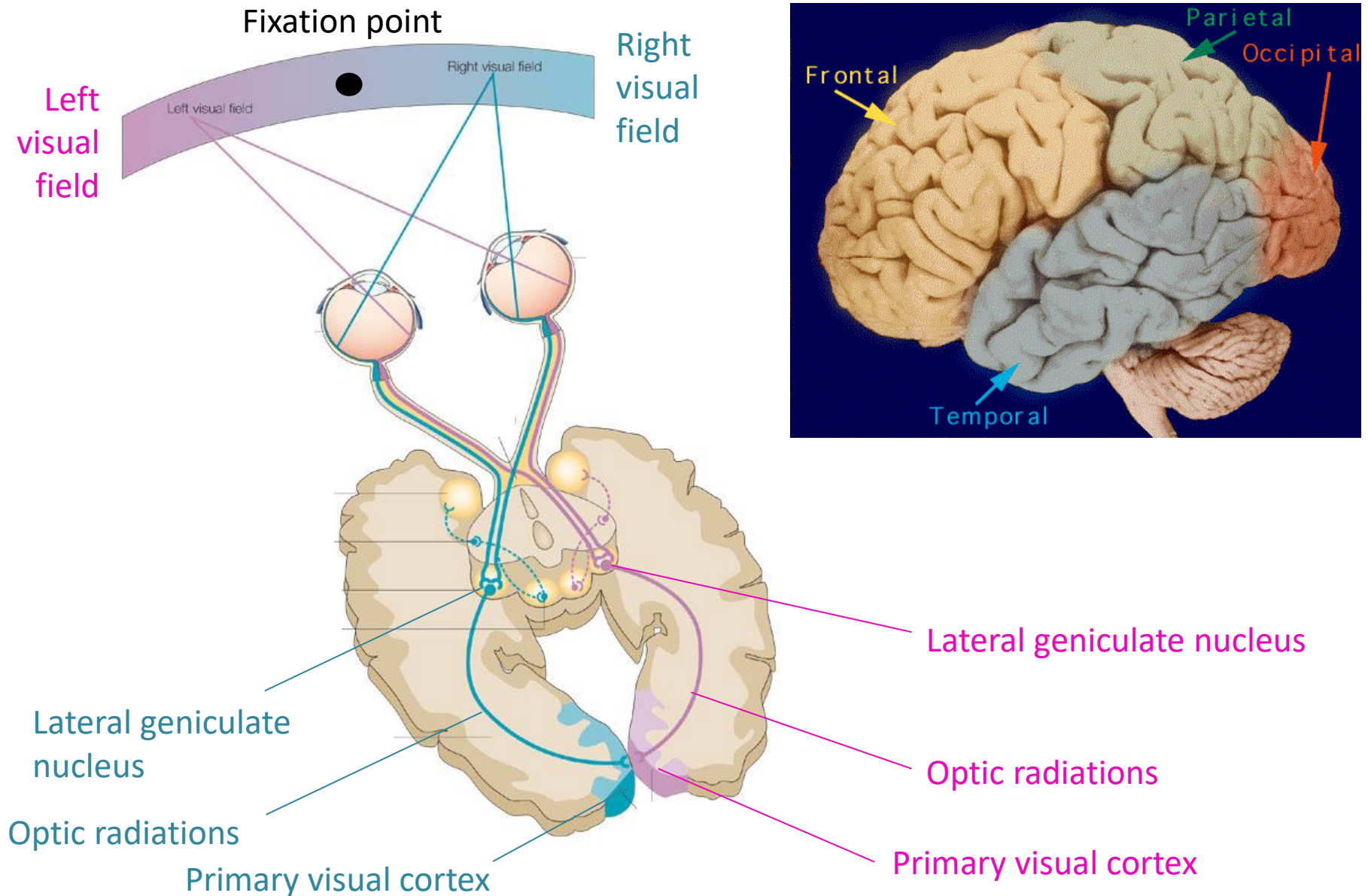
6 layers with
axons from:

Parvocellular
(P) neurons

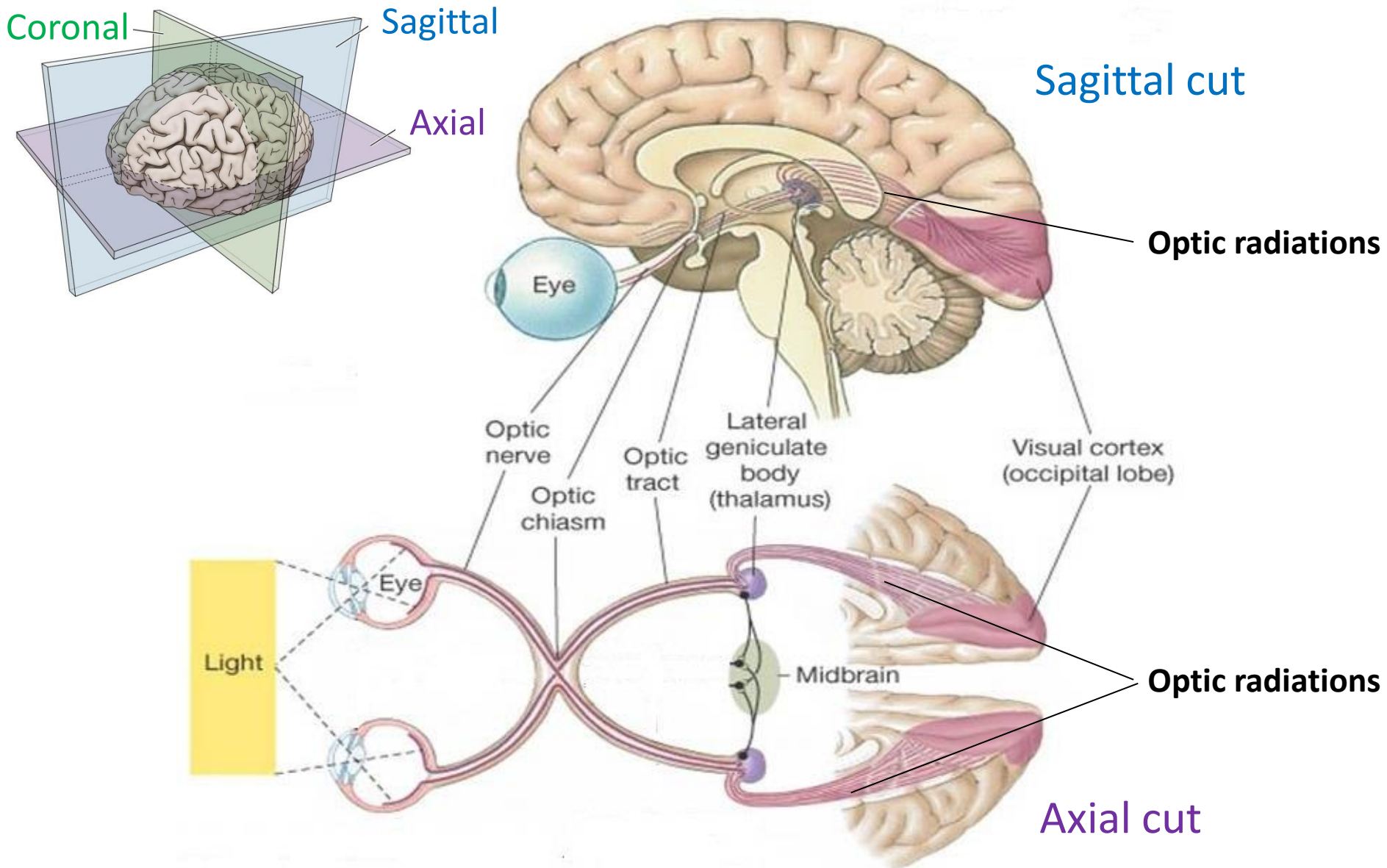
Magnocellular
(M) neurons

Koniocellular
(K) neurons
interlayered

Retino-geniculo-striate pathway



Retino-geniculo-striate pathway



Visual cortex

- Primary visual cortex
 - = V1 (visual area one)
 - = striate visual cortex
- Extrastriate visual areas
 - V2
 - V3
 - V4
 - V5 = MT (Middle Temporal area)
 - IT (Inferotemporal cortex)

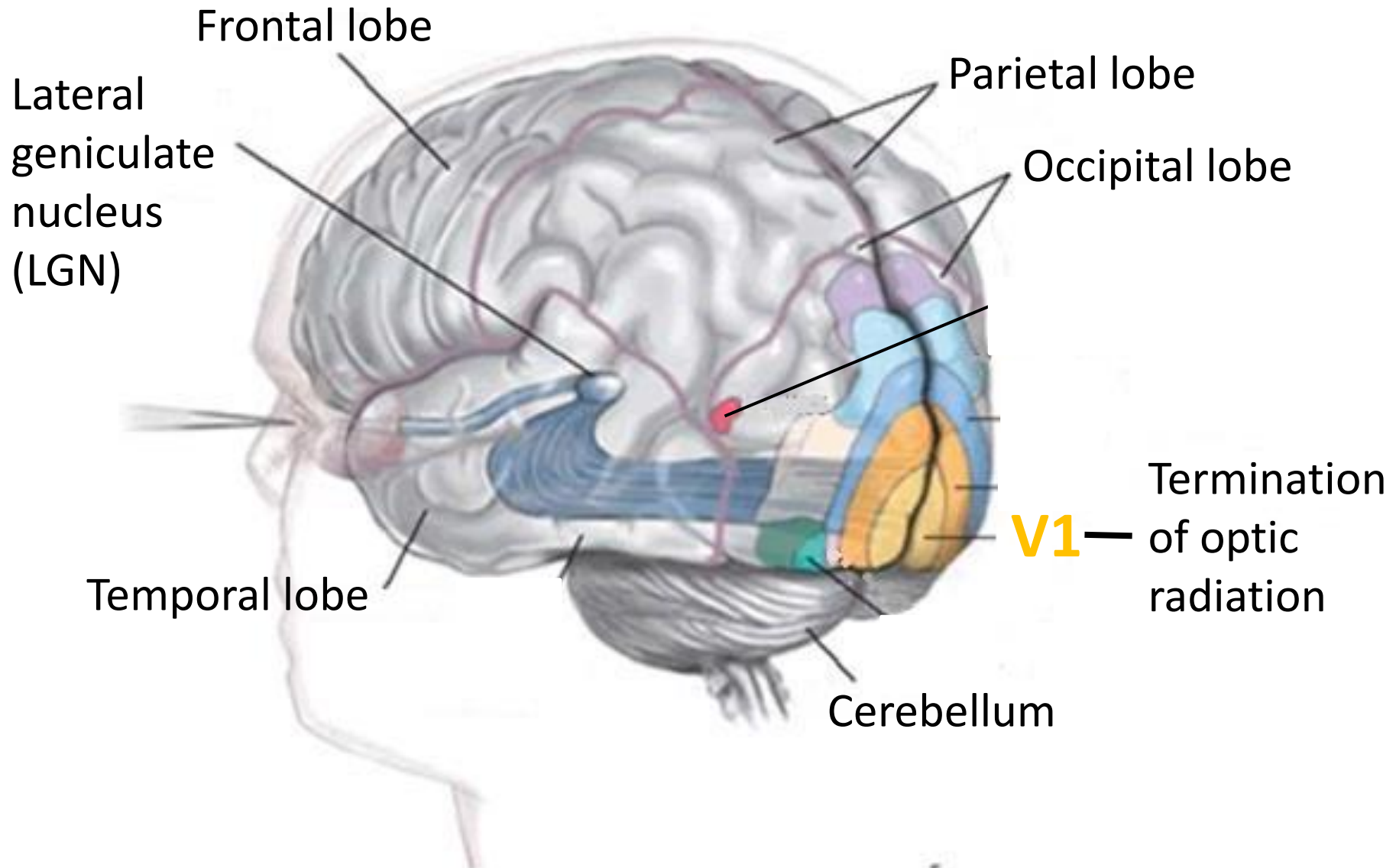


As visual information travels through the cortex, there is **ongoing neural convergence**:

The neurons' **receptive fields increase**.

Visual information gets **more complex** with ongoing integration.

Visual cortex

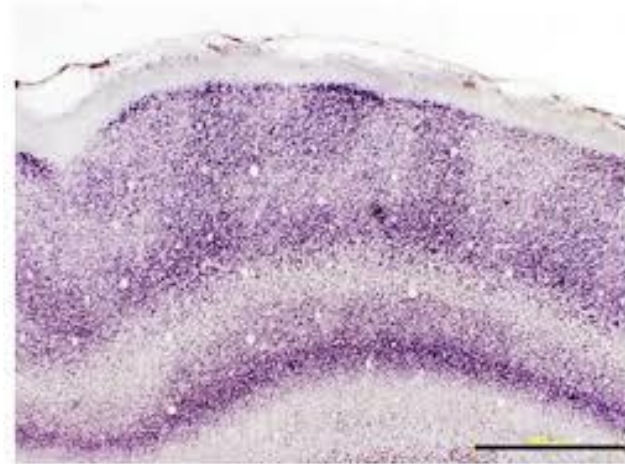
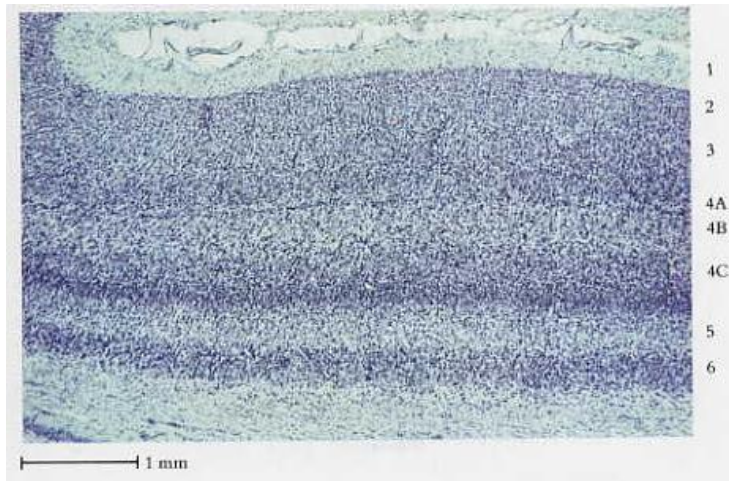


Visual cortex

- V1

- Striate cortex (stripy appearance)

Magno and parvo layers (from LGN and retina) are preserved in V1 input layers, but then merge subsequently.



Functions of V1 layers: David Hubel, “Eye, Brain, and Vision”: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/http://clinic.medlink.org/wp-content/uploads/2019/09/Eye_Brain_and_Vision.pdf

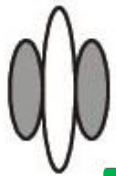
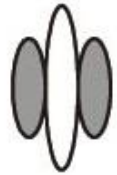
Visual cortex

- V1

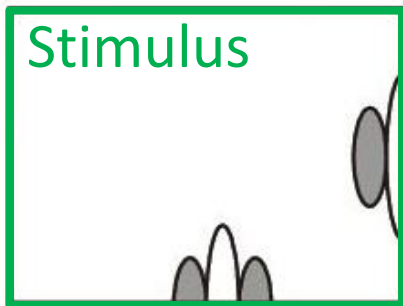
- Some V1 neurons are **orientation selective**.

They have **elongated receptive fields** to capture edges in a particular orientation.

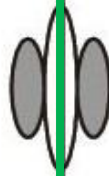
e.g., Receptive field that prefers vertical orientation.



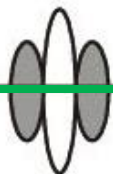
Receptive field not stimulated, **no response**



Stimulus



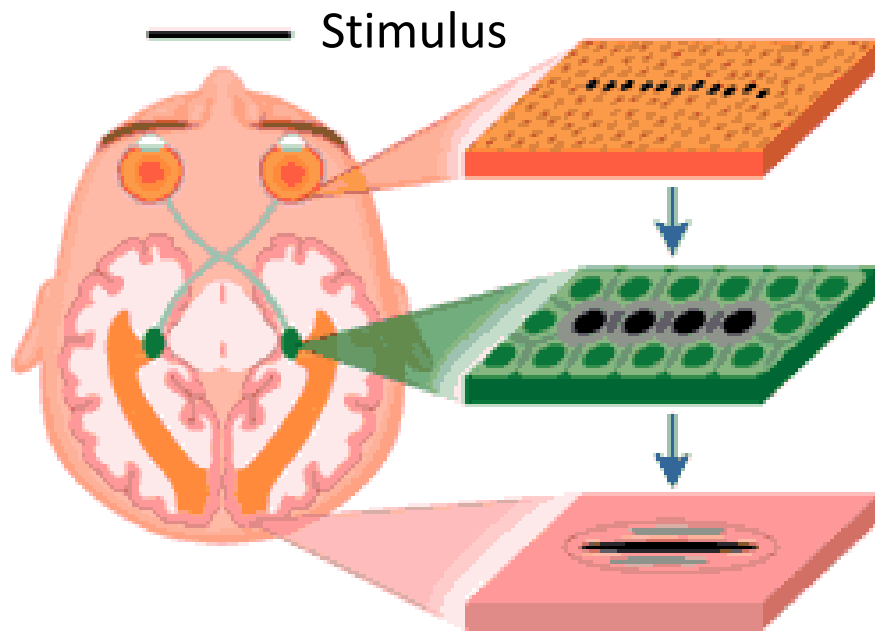
Receptive field stimulated with preferred orientation, **response**



Receptive field stimulated with non-preferred orientation, **no response**

Visual cortex

- V1
 - Some V1 neurons are **orientation selective**.



Ganglion cells in retina:
Centre-surround receptive fields.

Lateral geniculate nucleus (**LGN**):
Centre-surround receptive fields.

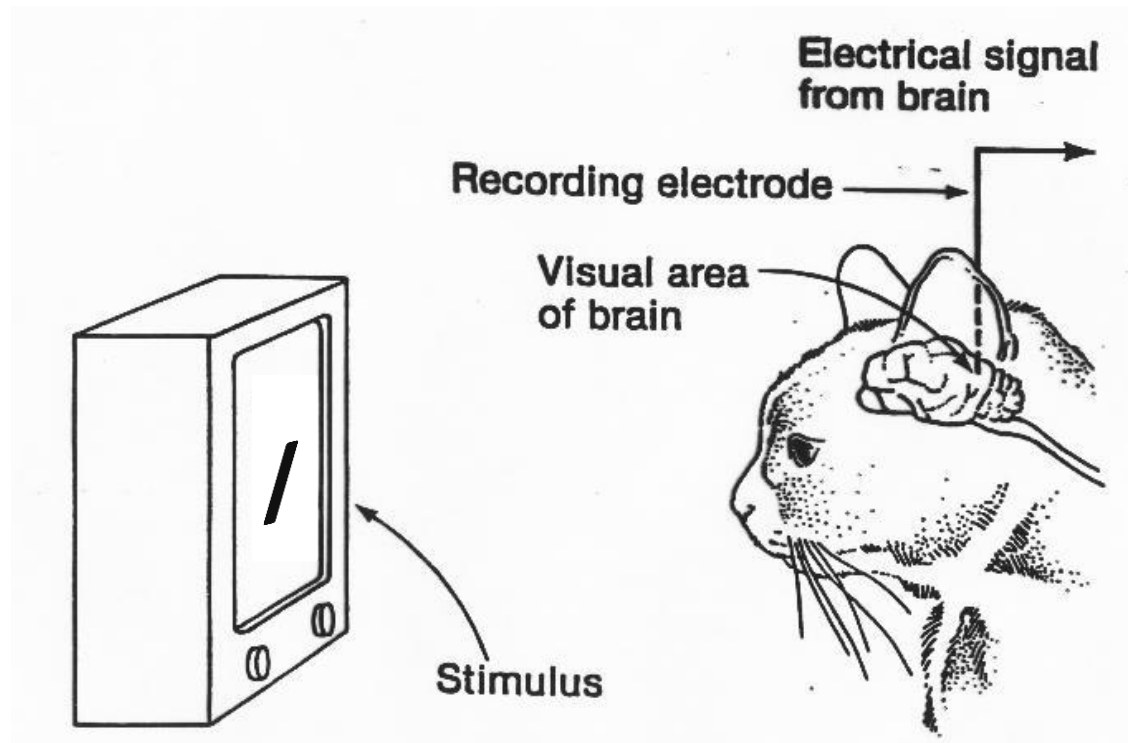
Primary visual cortex (**V1**):
Simple cortical cells, side-by-side
receptive fields.

Visual cortex

- (V1) single cell recording from Hubel & Wiesel, 1959

Video clip:

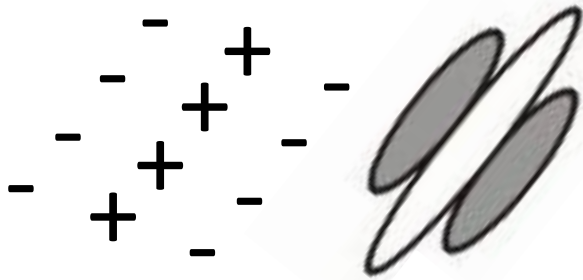
<https://www.youtube.com/watch?v=8VdFf3egwfg>



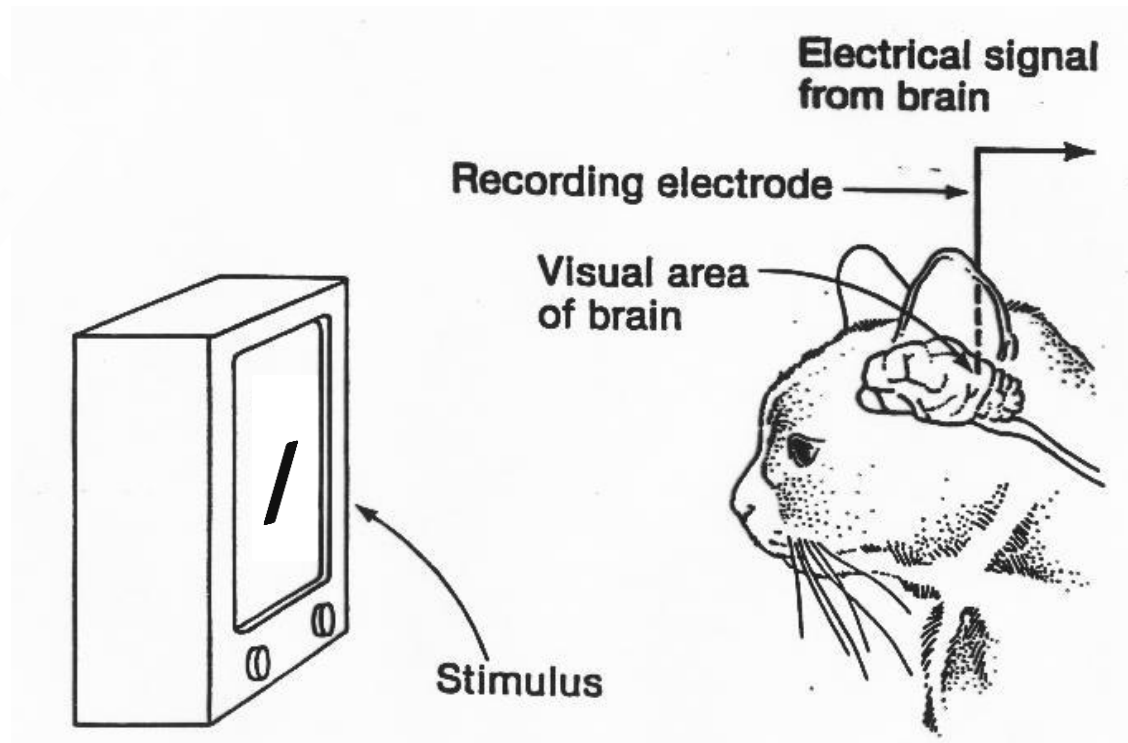
Hubel, D.H., & Wiesel, T.N. (1959). Receptive fields of single neurons in the cat's striate cortex. *Journal of Physiology*, 148, 574-591.

Visual cortex

- (V1) single cell recording from Hubel & Wiesel, 1959



Simple cortical cell with **side-by-side receptive field** (central excitatory area, inhibitory sides).



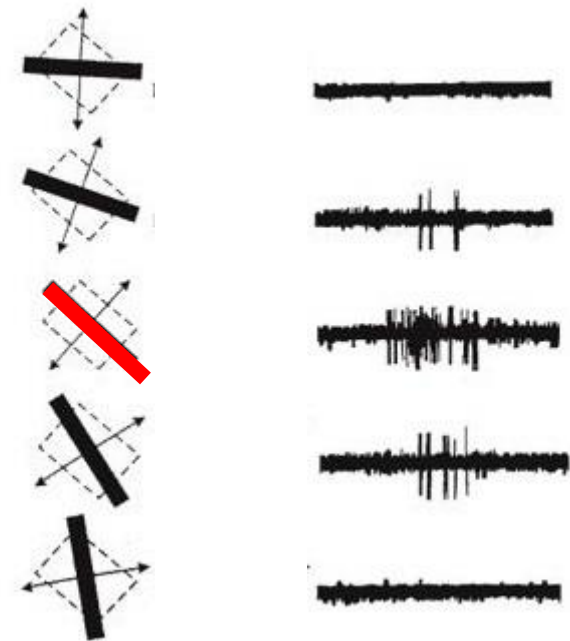
Watch Nobel Prize winner David Hubel lecture:
<https://www.youtube.com/watch?v=Gv6Edl-pidA>

Visual cortex

- V1
 - Some V1 neurons are **orientation selective**.
i.e., They respond when a stimulus in their receptive field matches their preferred orientation (Hubel & Wiesel, 1959).

Dashed rectangles
= neuron's receptive field.

The less the stimulus
matches the preferred
orientation, the lower is the
neuron's firing rate.



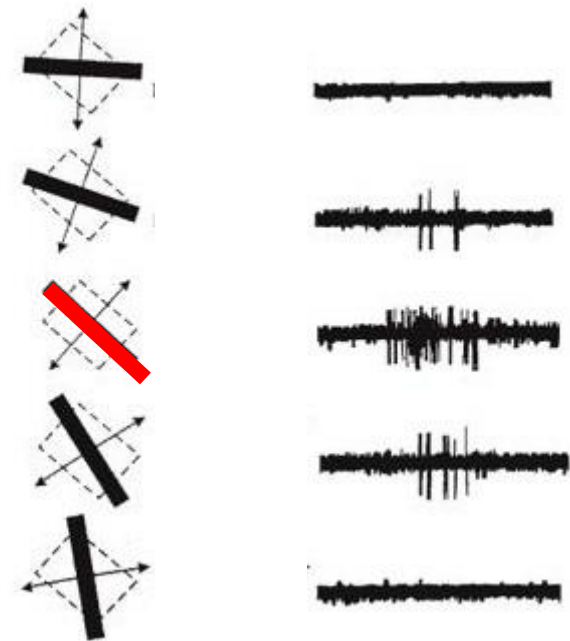
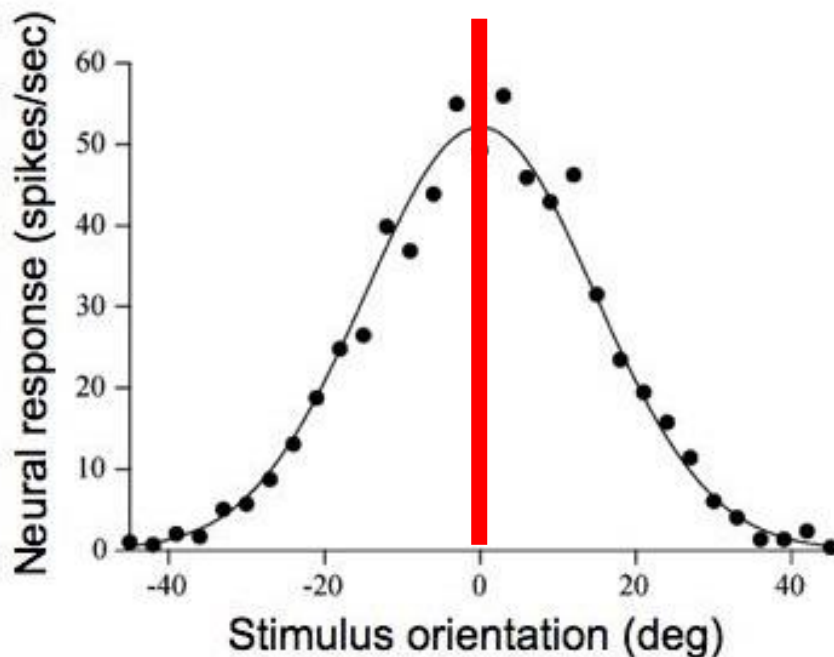
Stimulus Neuron's firing rate

Visual cortex

- V1

- Some V1 neurons are **orientation selective**.

i.e., They respond when a stimulus in their receptive field matches their preferred orientation (Hubel & Wiesel, 1959).



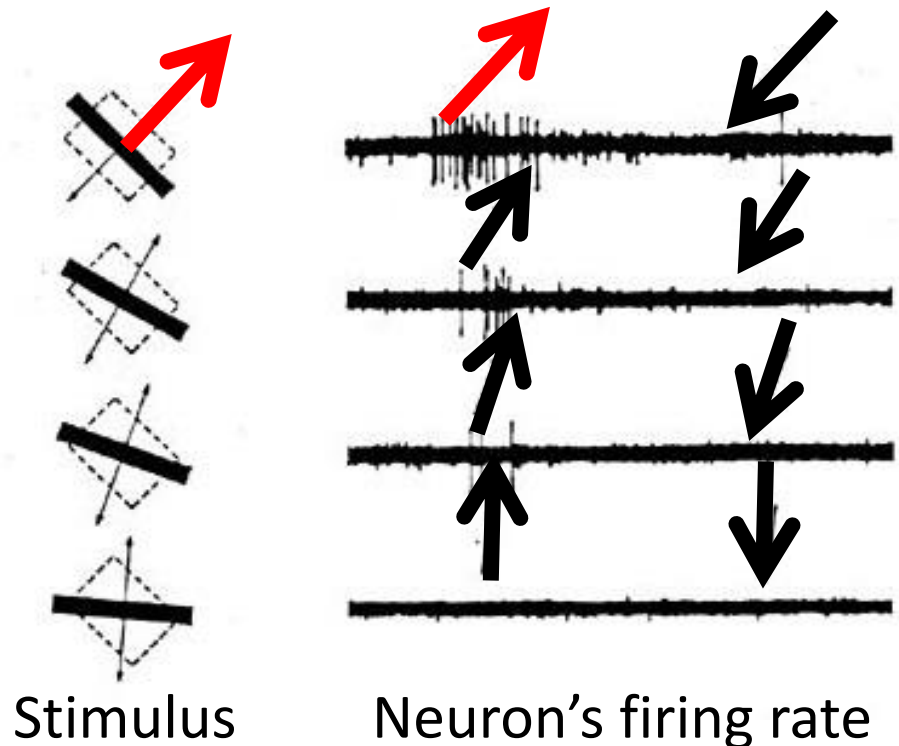
Stimulus Neuron's firing rate

Visual cortex

- V1
 - Other V1 neurons are **motion direction selective**.
i.e., They respond when a stimulus in their receptive field matches their preferred motion direction (Hubel & Wiesel, 1959).

Dashed rectangles
= neuron's receptive field

The less the stimulus
matches the preferred
motion direction, the lower
is the neuron's firing rate.



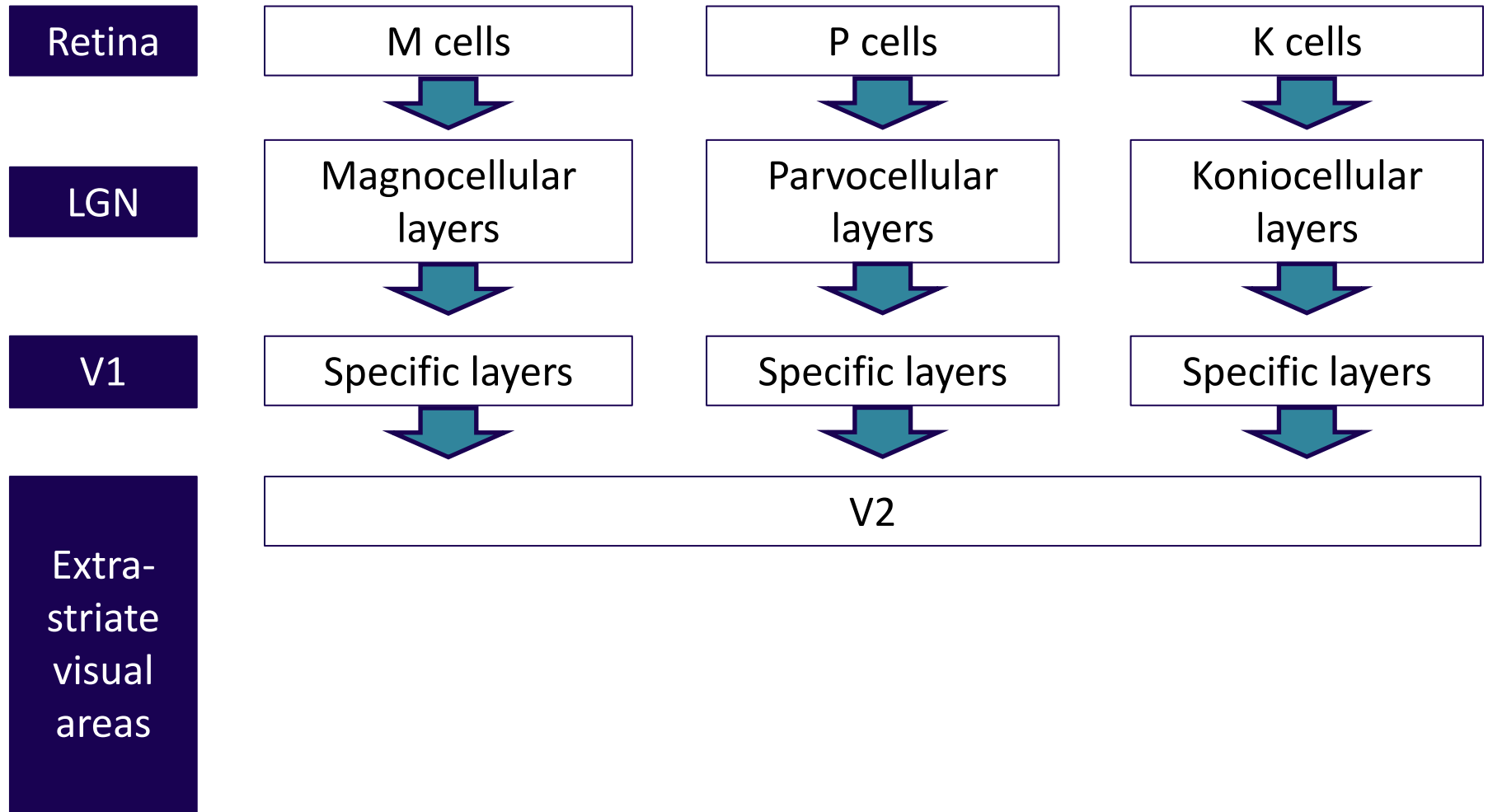
Visual cortex

- V1
 - Other V1 neurons are **selective for colour and brightness**.
(circular receptive fields; see lecture on colour perception)

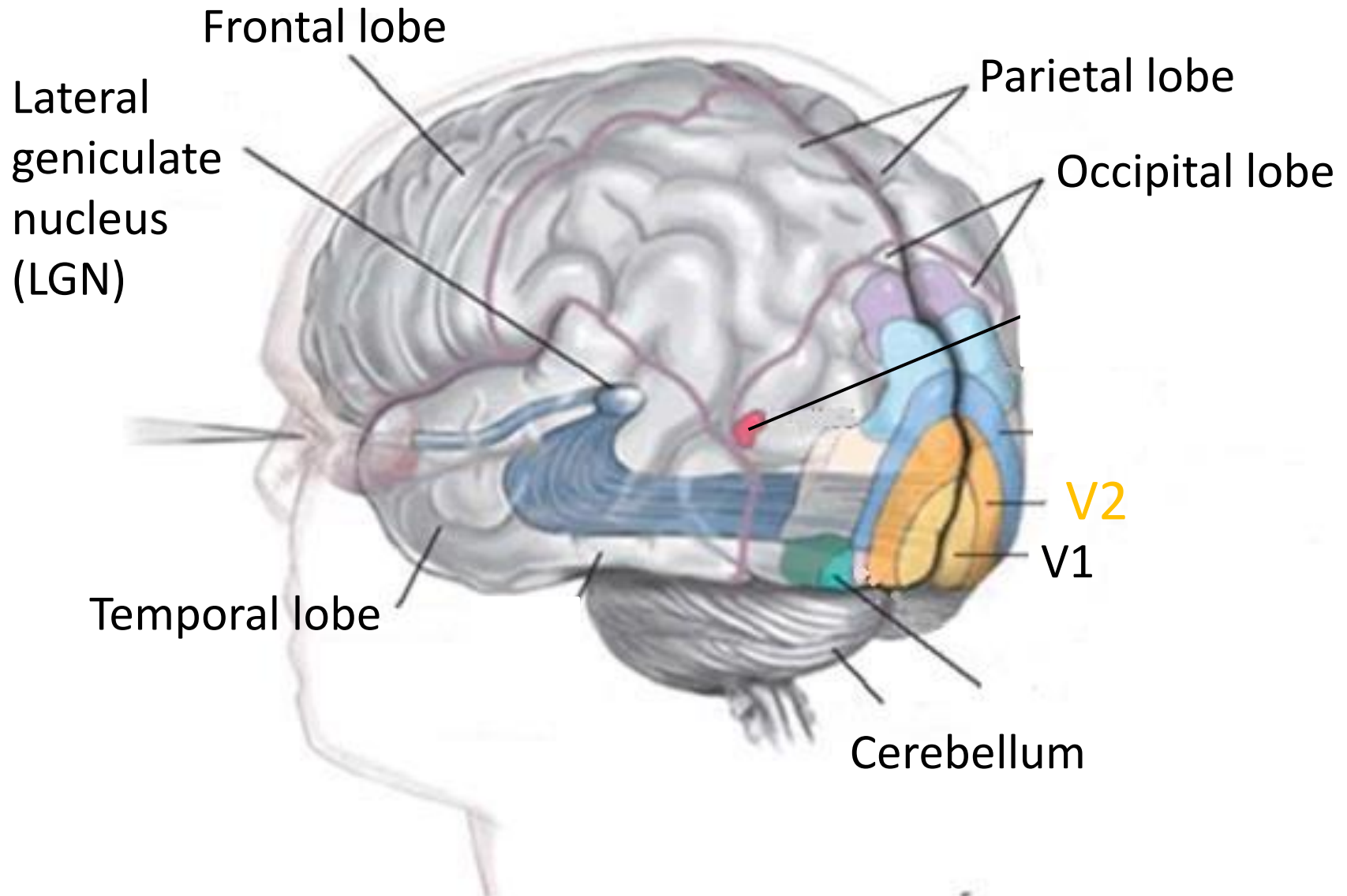
If you want to read a classic paper on visual cortex neurons and their receptive fields:

Livingstone , M., & Hubel, D. (1988). Segregation of form, color, movement, and depth: Anatomy, physiology, and perception. *Science*, 240, 740-749.

Visual cortex



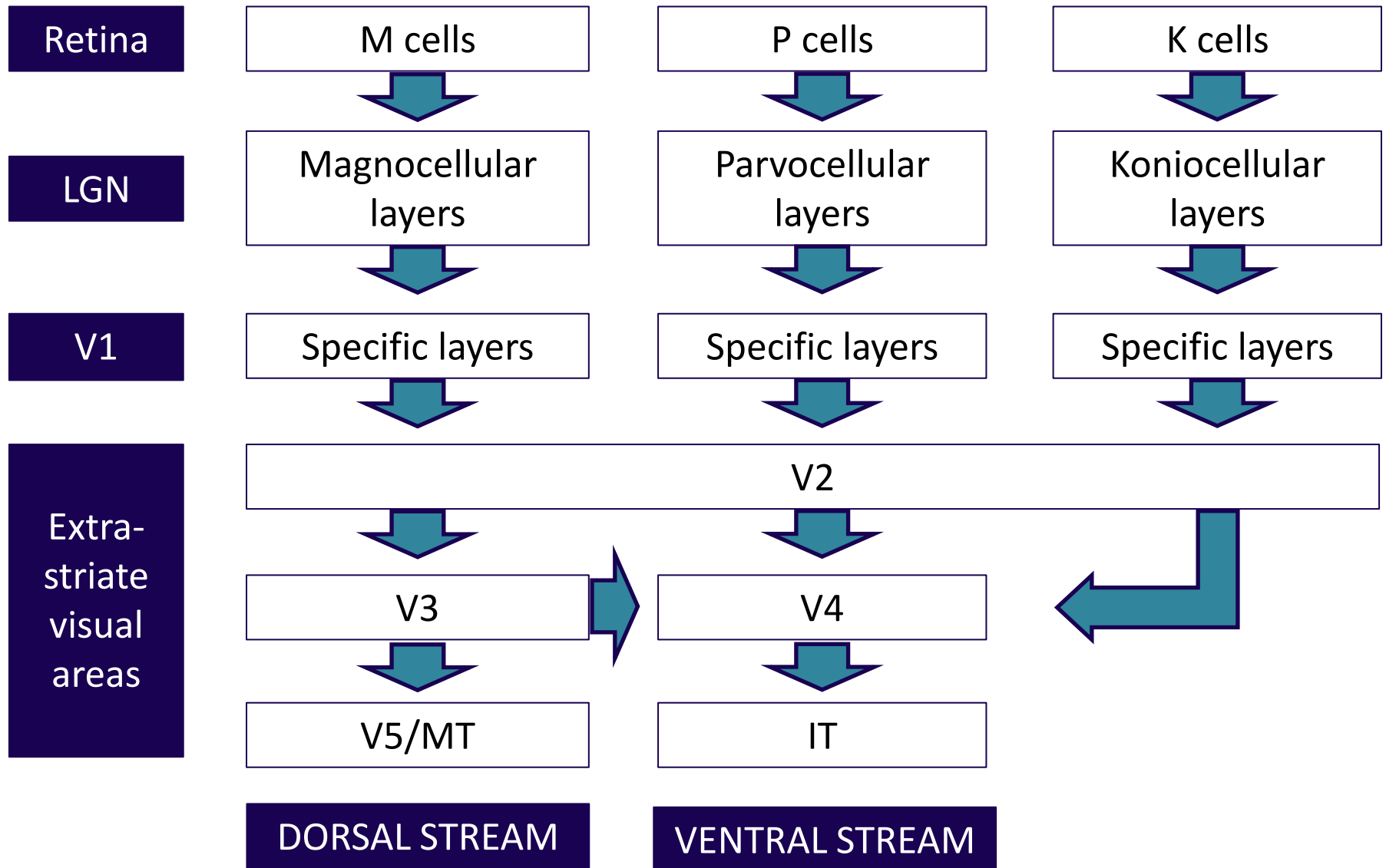
Visual cortex



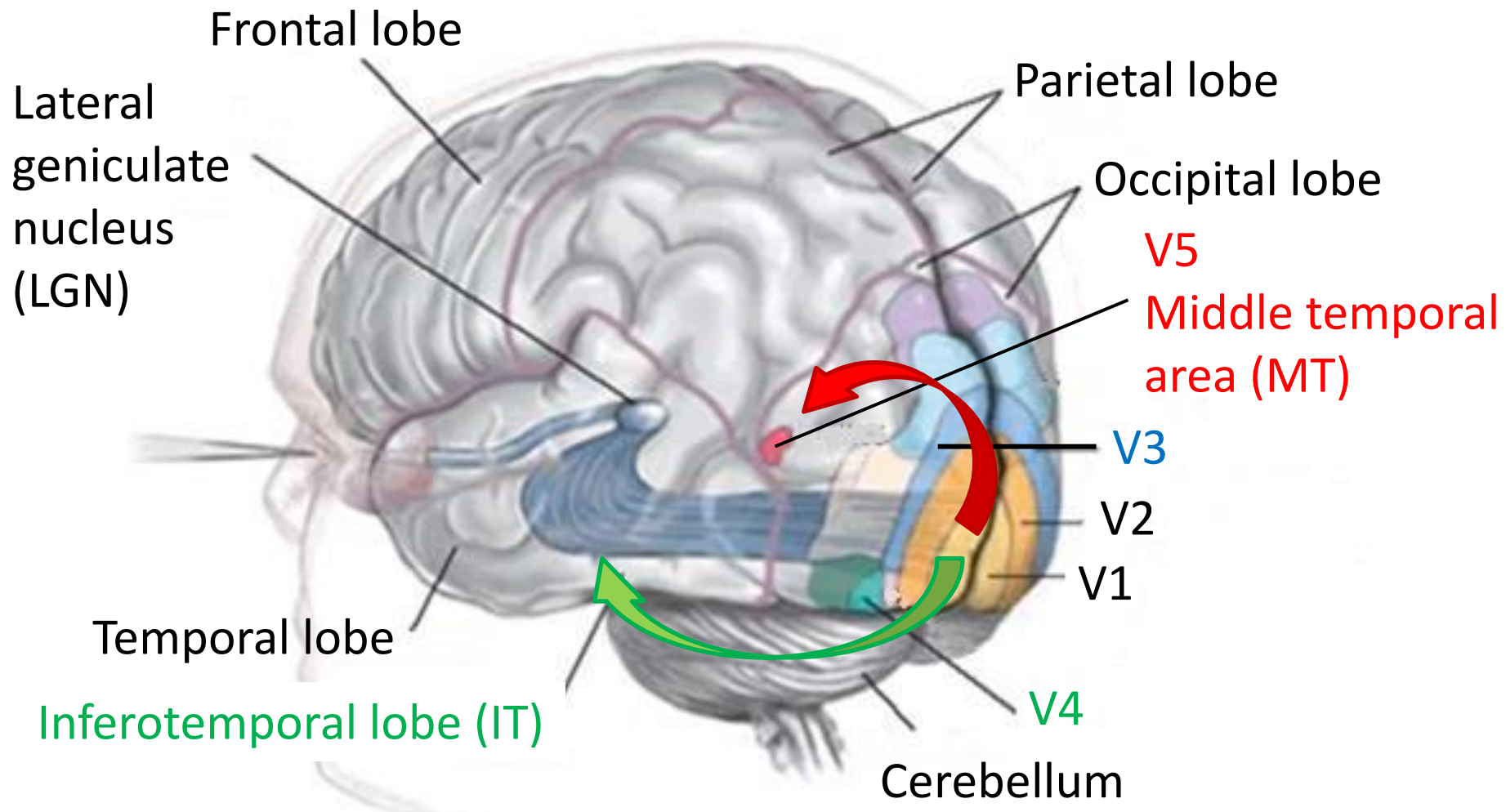
Visual cortex

- V2
 - V2 receptive fields are about **twice as large** as V1 receptive fields.
 - As V1, V2 neurons respond to **basic stimulus features** such as
orientation, motion direction, brightness, ...
 - Additionally, V2 neurons respond to **more complex features** such as
length, angles, arcs, shapes, components of natural scenes, texture, ...

Visual cortex



Visual cortex



Temporal (ventral) stream: V1/2/4/IT (ventrum = belly)

Parietal (dorsal) stream: V1/2/3/MT (dorsum = back)

Visual cortex

- Evidence for two separate streams
 - Ungerleider & Mishkin, 1982

Parietal (dorsal) stream processes object locations.
= **WHERE?**

Temporal (ventral) stream processes object identities.
= **WHAT?**

Ungerleider, L.G., & Mishkin, M. (1982). Two cortical visual systems. In D. J. Ingle, M.A. Goodale, R.J.W. Mansfield (eds), *Analysis of visual behaviour*. Boston, MA: MIT Press, pp. 540-586.

Visual cortex

- Evidence for two separate streams

- Ungerleider & Mishkin, 1982

Monkeys with parietal ('where') lesions could distinguish a cube from a triangle (**object discrimination**), while monkeys with temporal ('what') lesions could not.

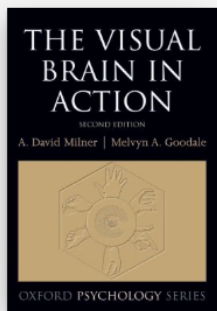
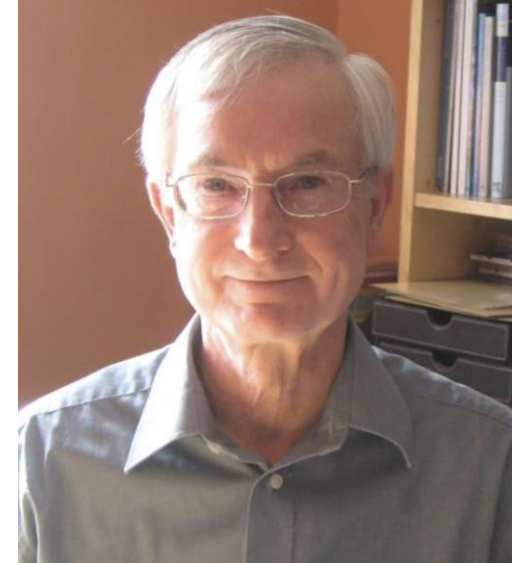
Monkeys with temporal ('what') lesions could learn the position of an object (**location discrimination**), while monkeys with parietal ('where') lesions could not.

→ Perfect **double dissociation**! Damage of different brain areas leads to the opposite pattern of impairments; i.e., those brain areas have separable functions.

Visual cortex

- Evidence for two separate streams
 - Prof David Milner

**Emeritus Professor in Psychology
at Durham!!**



BOOK

The Visual Brain in Action

Milner, David ; Goodale, Mel; Goodale, Mel
2006

[Available at Bill Bryson Library Level 3 \(612.8255 MIL\) >](#)

[Available Online >](#)

[Chapters of this book \(16\) >](#)

Visual cortex

- V3 and MT in the parietal/dorsal (where) stream
 - Receptive fields are five times larger in V3 and eight times larger in MT than in V1.

Integrated information over a large area of the retina
i.e., perfect for **motion perception** (where is an object; see lecture on motion perception).

Visual cortex

- V4 in the temporal/ventral (what) stream
 - Receptive fields are five times larger in V4 than in V1.
 - V4 neurons respond to **object-defining features** (what is an object) such as
colour, orientation, complex shapes, texture, ...

If you want to be surprised by the stimulus complexity V4 neurons respond to:
Kobatake, E. & Tanaka, K. (1994). Neuronal selectivities to complex object features in the ventral visual pathway of the macaque cerebral cortex. *Journal of Neurophysiology*, 71, 856-867.

Visual cortex

- A little bit of **controversy**
 - How useful is the separation between two visual processing streams?
 - The **dorsal and ventral stream** seem to **interact** heavily, as for example
 - reaching and grasping is more precise towards familiar objects.
 - i.e., directional movements, controlled by the dorsal stream, are influenced by object recognition in the ventral stream.

If you like controversy: de Haan, E.H., & Cowey, A. (2011). On the usefulness of 'what' and 'where' pathways in vision. *Trends in Cognitive Sciences*, 15, 460-466

Summary

- We have looked at
 - (M, P, and K) **ganglion cells** in the retina and their receptive fields (on/off centre/surround).
 - The **retino-geniculo-striate pathway**.
Lateralisation of visual processing.
 - Cortical visual processing starting at **V1** (simple cortical cells, side-by-side receptive fields).
Neurons are selective for **basic stimulus features** (e.g., orientation, motion direction, colour, luminance).

Summary

- We have looked at
 - Two visual processing streams of **cortical processing in extrastriate areas** (increasing receptive fields, increasing complexity of visual information).
 - Temporal (ventral) stream** (V1/2/4/IT): **What** stream (object identification, neurons respond to object-defining features).
 - Parietal (dorsal) stream** (V1/2/3/MT): **Where** stream (object localisation, neurons respond to motion).

Test yourself – learning goals

- After reviewing this lecture, you should be able to:
 - Explain what a receptive field is, what types there are, and what they do for visual processing.
 - Name 3 types of ganglion cells and LGN layers.
 - Draw a rough draft of the retino-geniculo-striate pathway and name its most important landmarks.
 - On the basis of your drawing, explain which parts of the visual field are represented in which hemisphere.
 - Explain when an orientation or motion direction selective neuron in V1 fires and when not.
 - Describe neural convergence and what it means with respect to the size of receptive fields.
 - Name the two cortical visual pathways, label their landmarks, and describe their outcome in terms of stimulus processing.

Test yourself – multiple choice

1. Which sentence is correct? Following the retino-geniculo-striate pathway, ...

- a) ... the right visual field is represented fully in the right striate cortex.
- b) ... the right visual field is represented temporally in the right lateral geniculate nucleus.
- c) ... the right visual field is represented nasally in the right eye.
- d) ... the right visual field is represented partially in the right optic chiasm.

Test yourself – multiple choice

2. From Hubel and Wiesel's (1959) single cell recordings we have learned that ...

- a) ... orientation-selective cells in V1 have elongated receptive field to capture the preferred motion direction of a visual stimulus.
- b) ... orientation-selective cells in V1 are simple cortical off-centre cells.
- c) ... orientation-selective cells in V1 have centre-surround receptive fields very similar to those in the lateral geniculate nucleus.
- d) ... orientation-selective cells in V1 have side-by-side receptive fields.

Test yourself – multiple choice

3. There are two cortical visual pathways.

- a) One projects to the temporal lobe and one to the parietal lobe.
- b) One projects to the inferotemporal lobe and one to the temporal lobe.
- c) One projects to the parietal lobe and one to the occipital lobe.
- d) One projects to the inferotemporal lobe and one to the occipital lobe.

See you on Tuesday!

